

Spatial Phylodiversity & Phyloendemism

JT Miller

University of Florida

FLORIDA MUSEUM

UF UNIVERSITY of FLORIDA

iDigBio
Integrated Digitized Biocollections

NSF

Background: How can we measure Biodiversity?

Presence/Absence Data

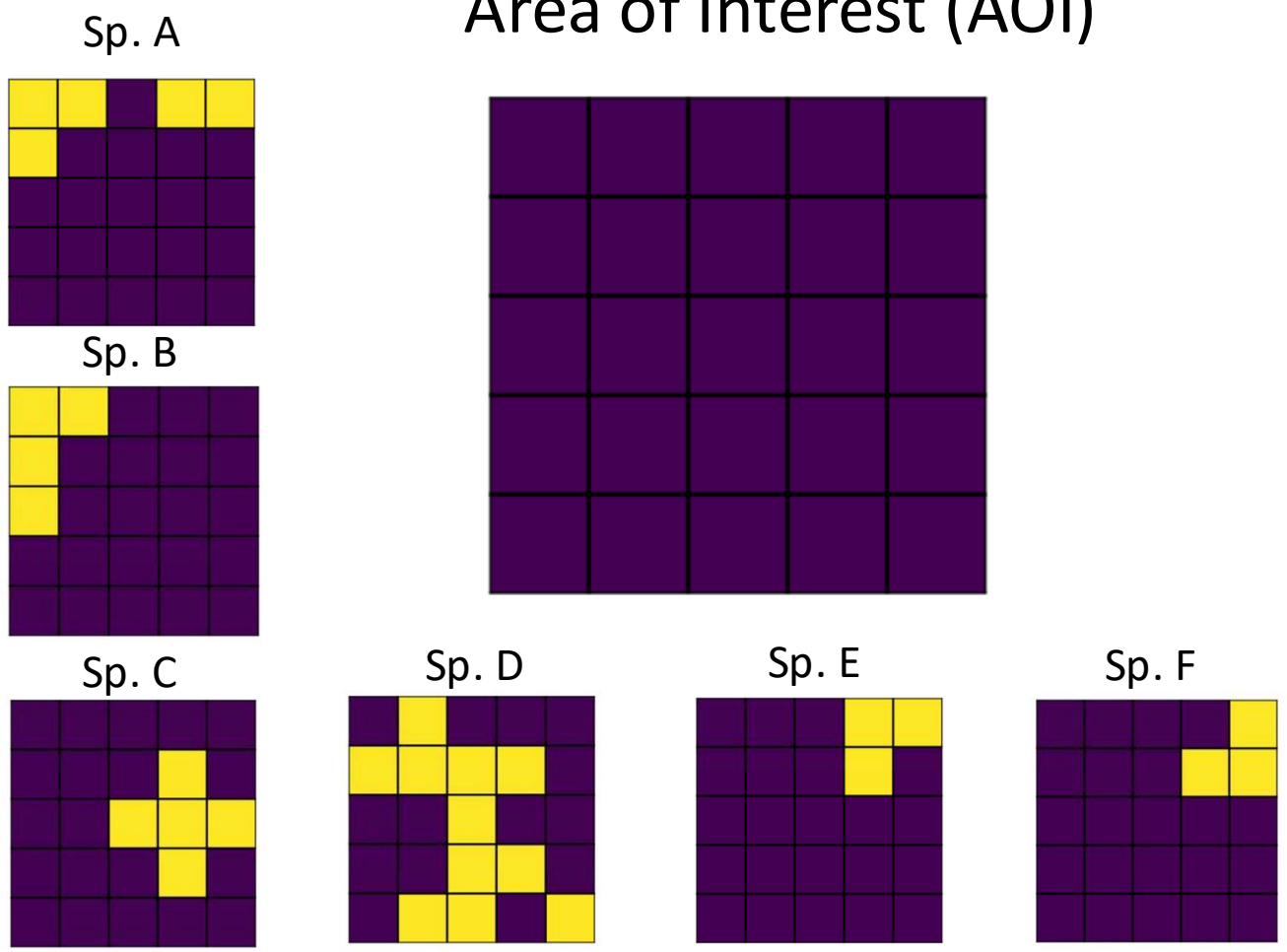
- Species Richness (SR)

What do you need?

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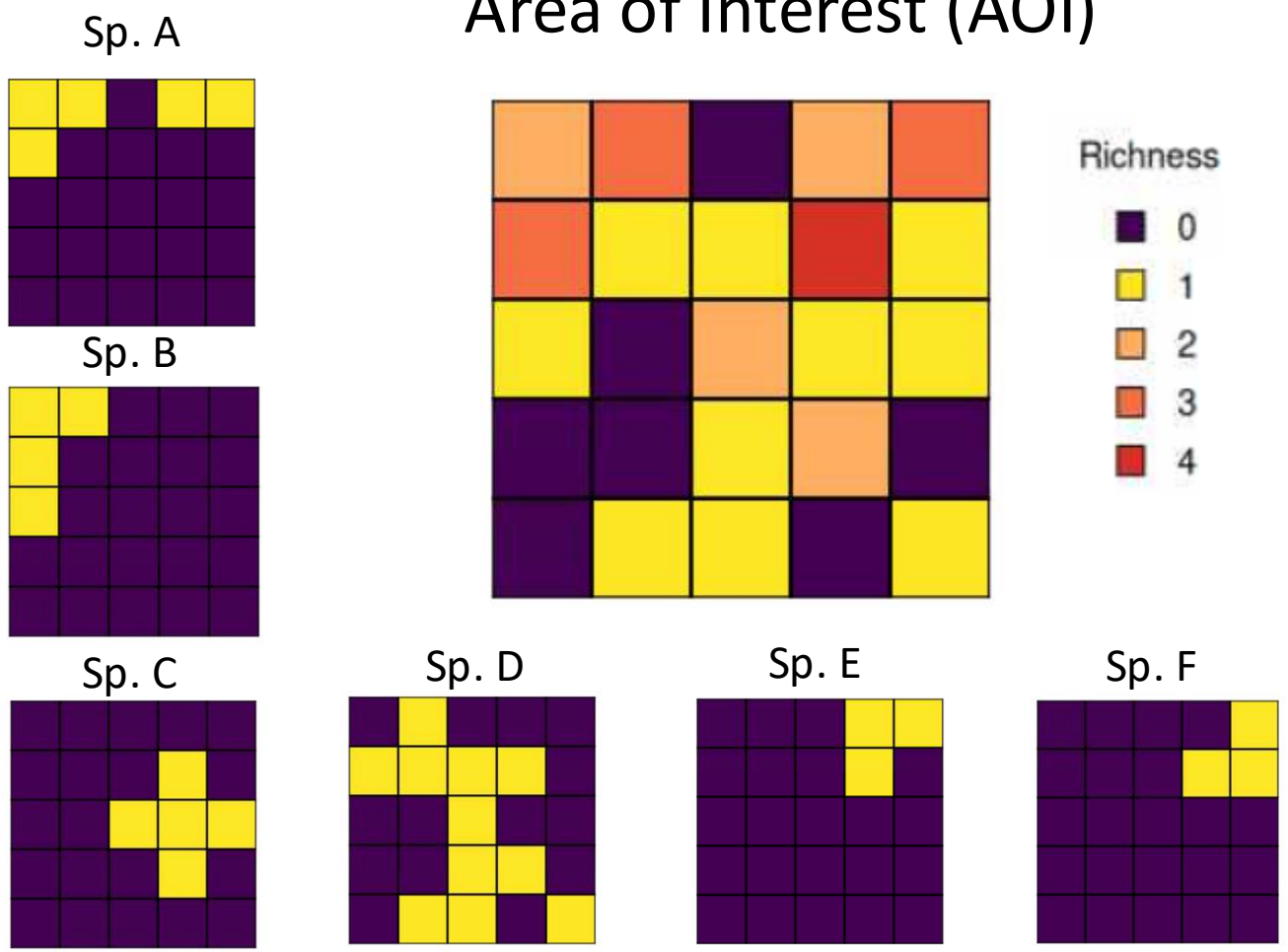
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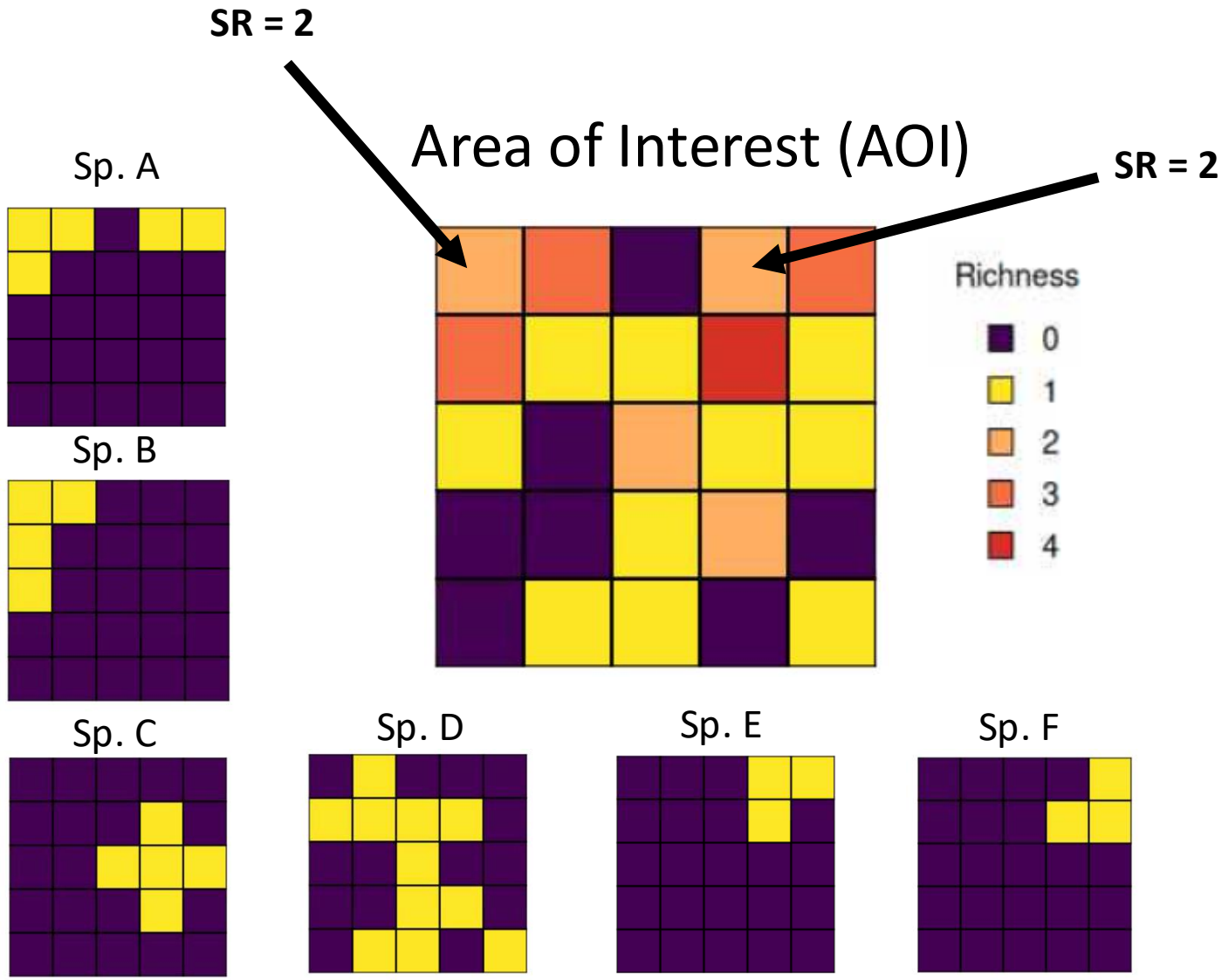
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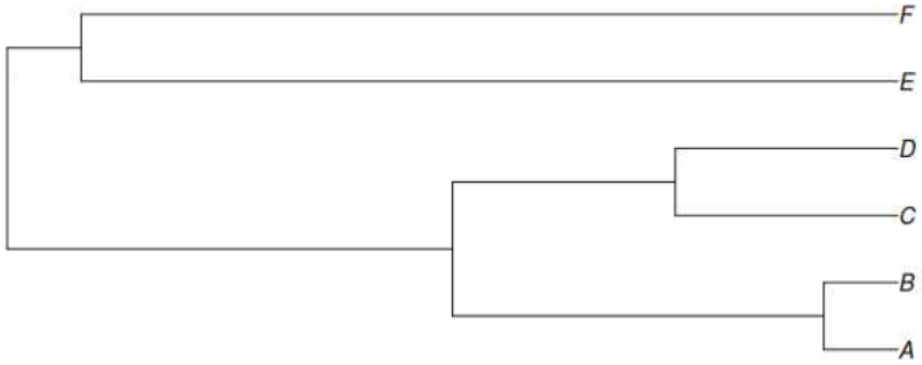
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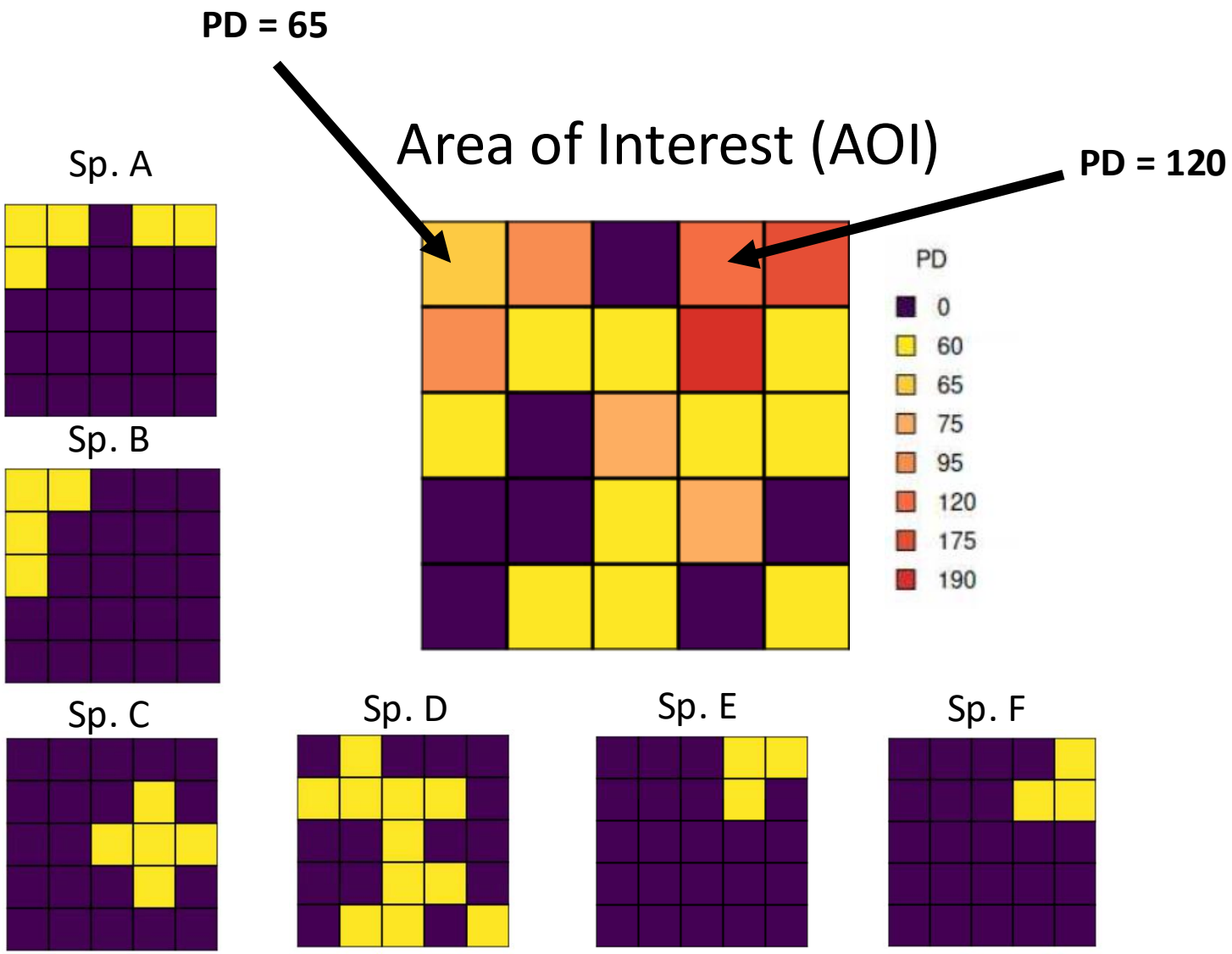
Presence/Absence Data

- **Phylogenetic Diversity (PD)**

What do you need?

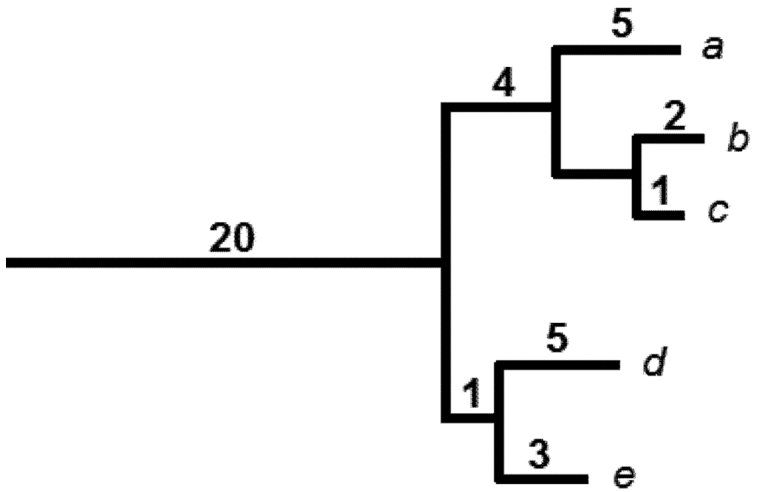
1. A study area / (AOI)
2. Community P/A data
3. A phylogeny





Background: What is Phylodiversity?

- How much of the phylogenetic tree/evolutionary history is represented in my sample?
  - Uses phylogenetic distance (PD)



Faith & Richards 2012

Biological Conservation 1992, **61**, 1–10

Conservation evaluation and phylogenetic diversity

Daniel P. Faith

Division of Wildlife and Ecology, CSIRO, PO Box 84, Lyneham, ACT, 2602, Australia

(Received 13 May 1991; revised version received 28 September 1991; accepted 11 November 1991)

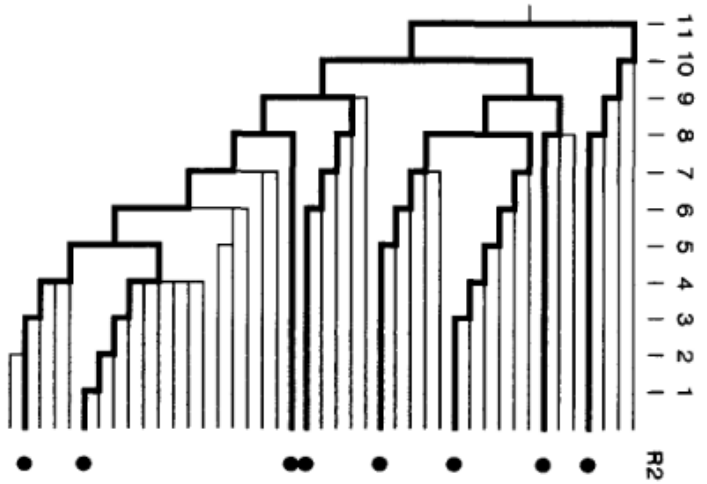
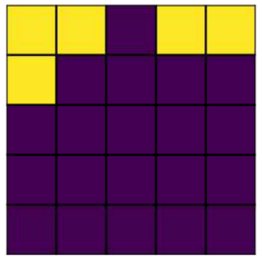


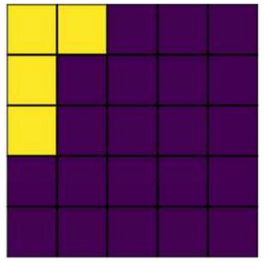
Fig. 3(b). The cladogram for bumble bee species, showing the minimum spanning path for the species found in R2. This reserve area shows highest phylogenetic diversity.

PD Calculation

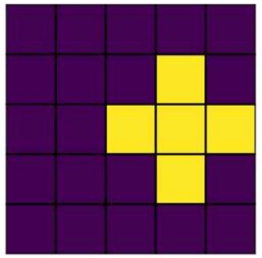
Sp. A



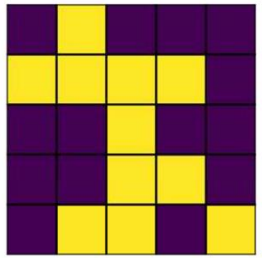
Sp. B



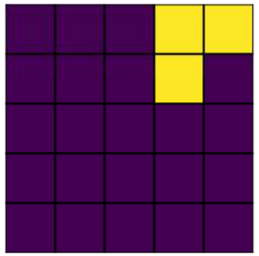
Sp. C



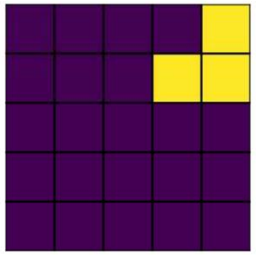
Sp. D



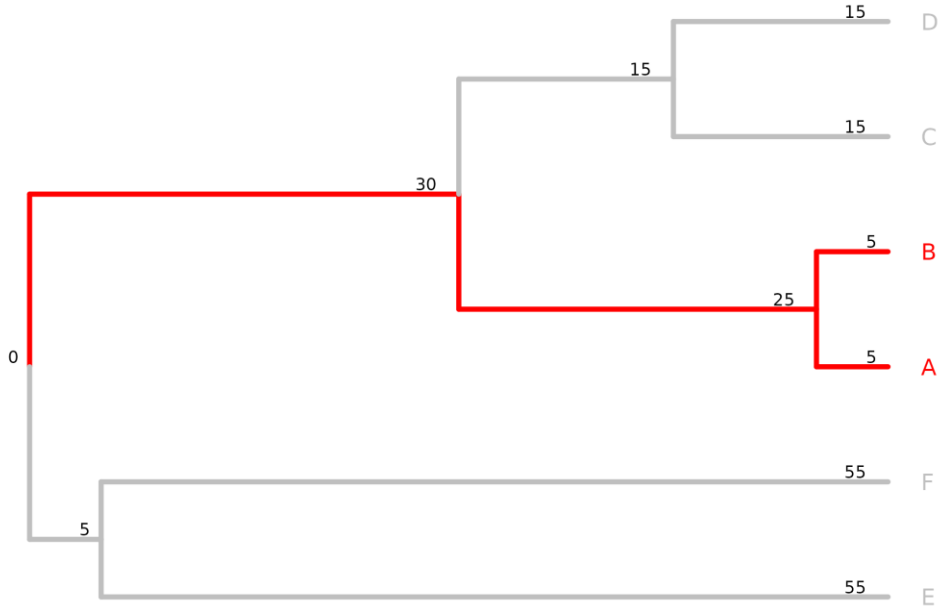
Sp. E



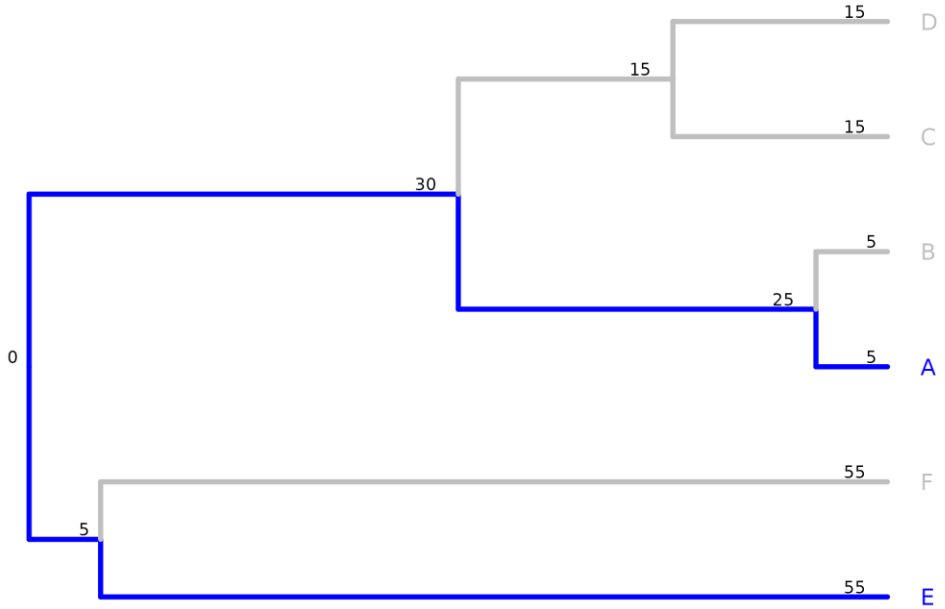
Sp. F



Cell 1 | row 1, col 1
Taxa: A, B
Species Richness: 2
PD: 65



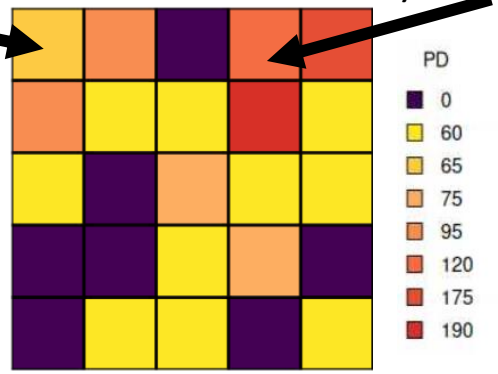
Cell 4 | row 1, col 4
Taxa: A, E
Species Richness: 2
PD: 120



PD = 65

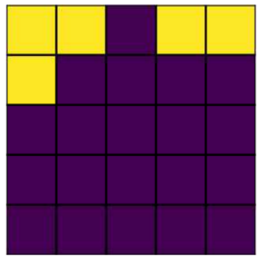
Area of Interest (AOI)

PD = 120

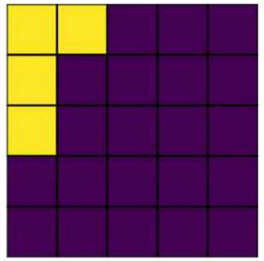


PD Calculation

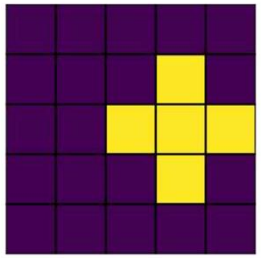
Sp. A



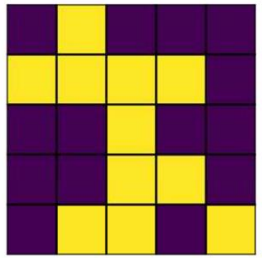
Sp. B



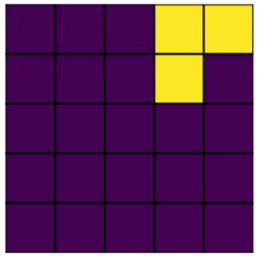
Sp. C



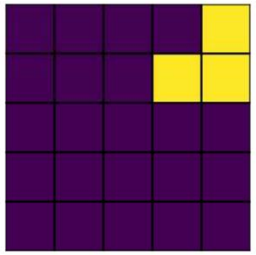
Sp. D



Sp. E

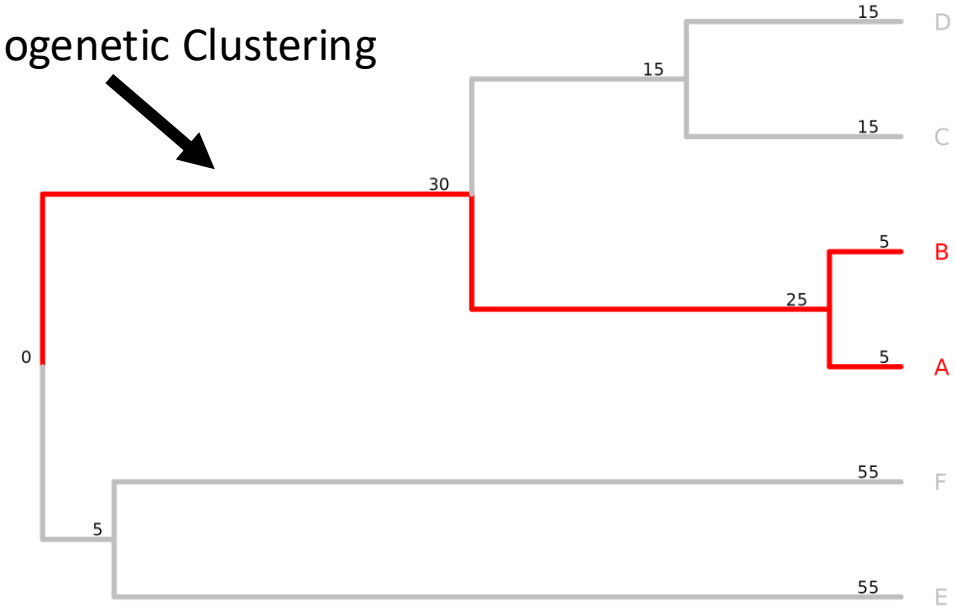


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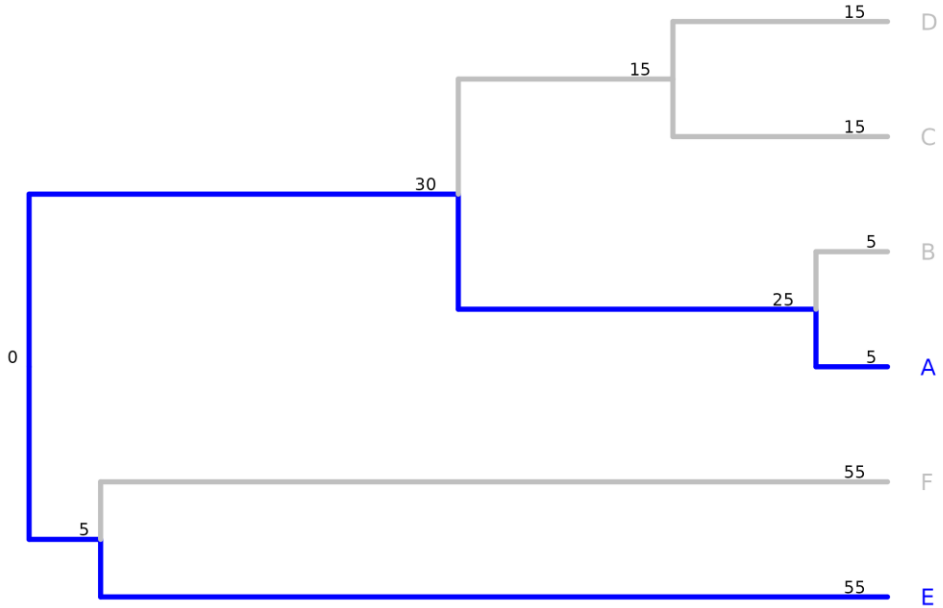


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Phylogenetic Clustering



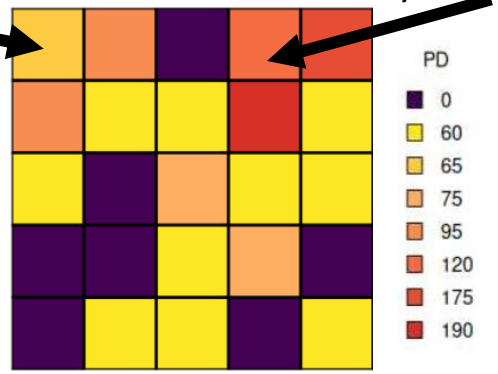
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Phylogenetic overdispersion

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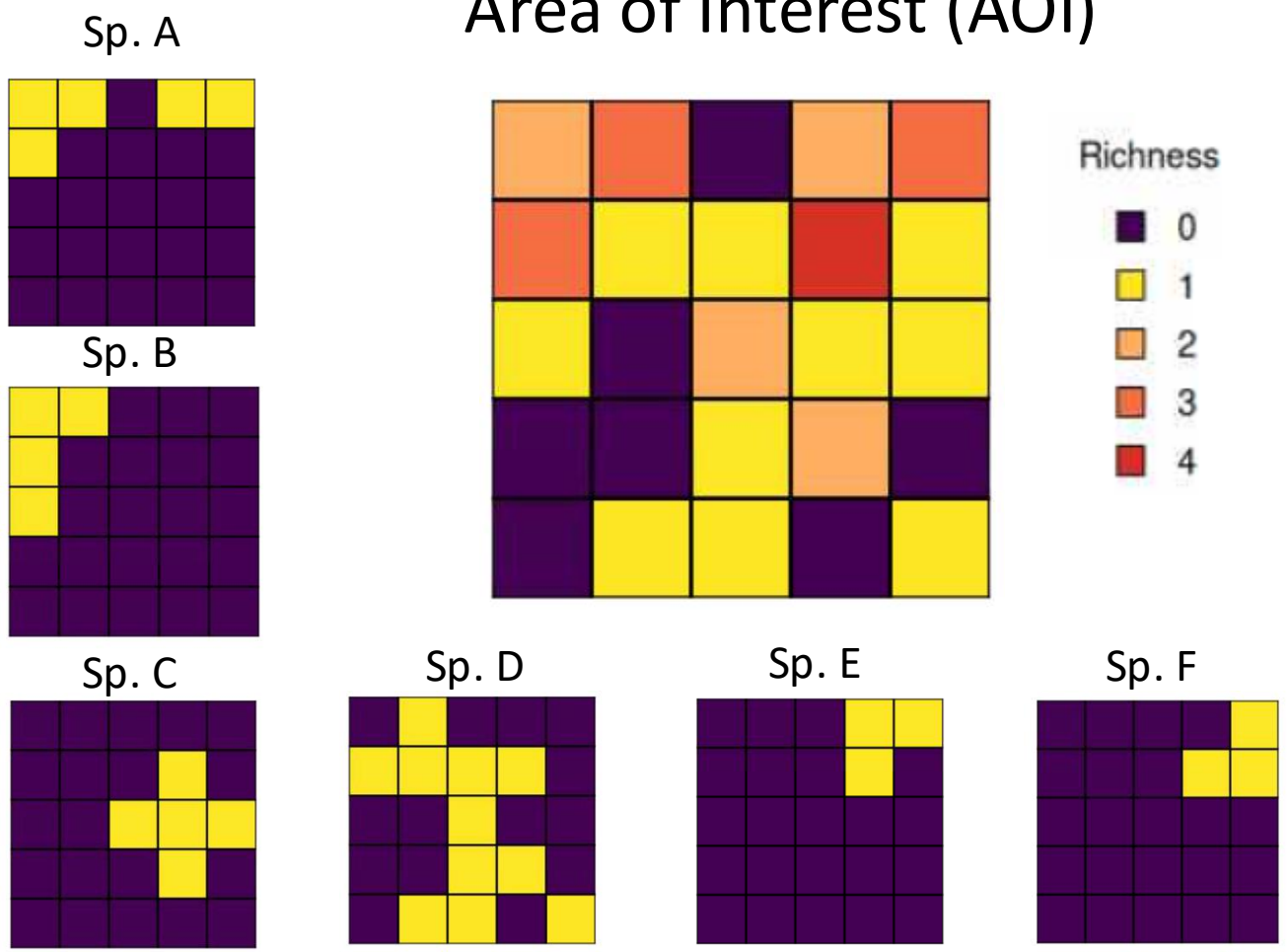
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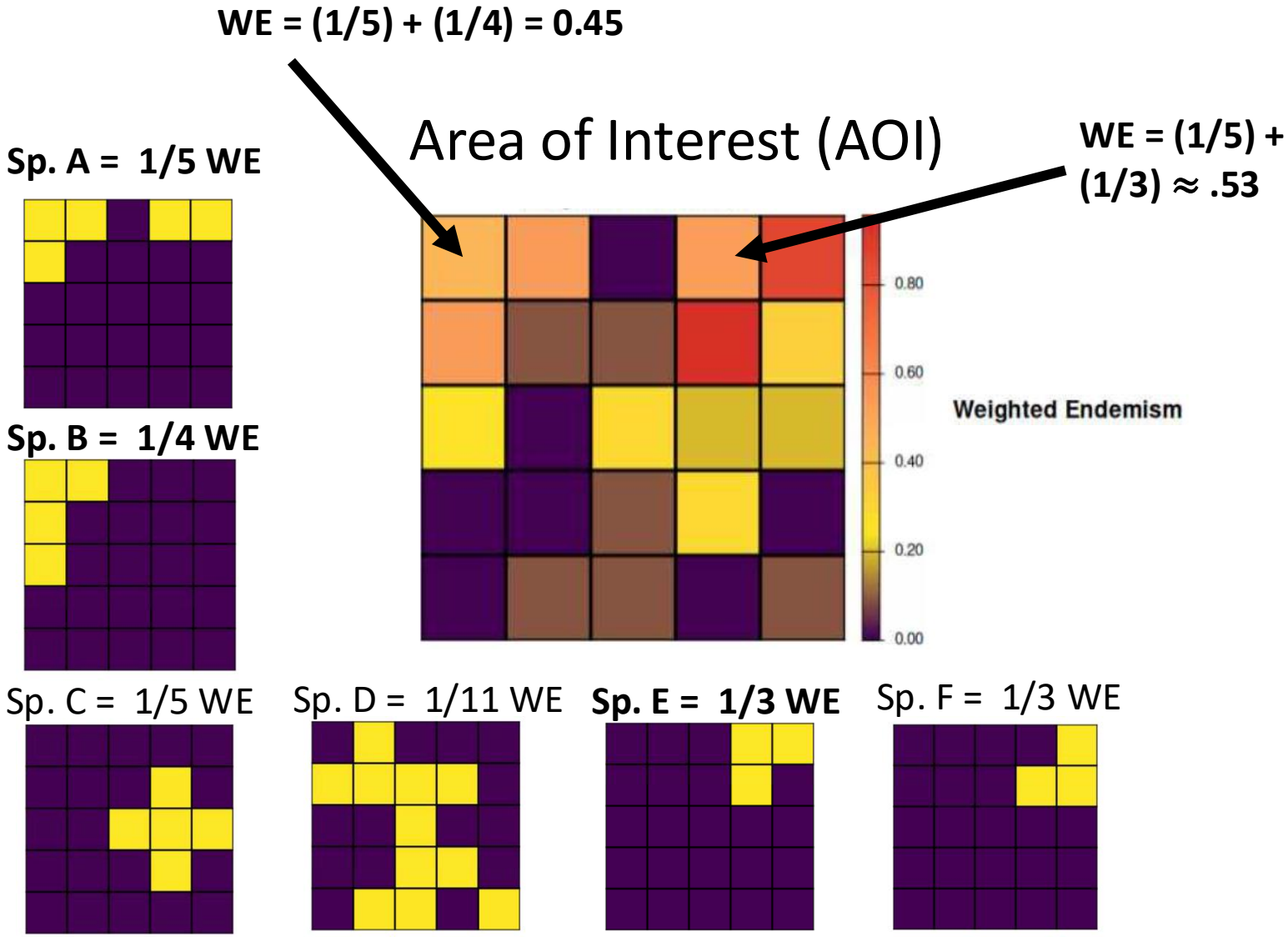
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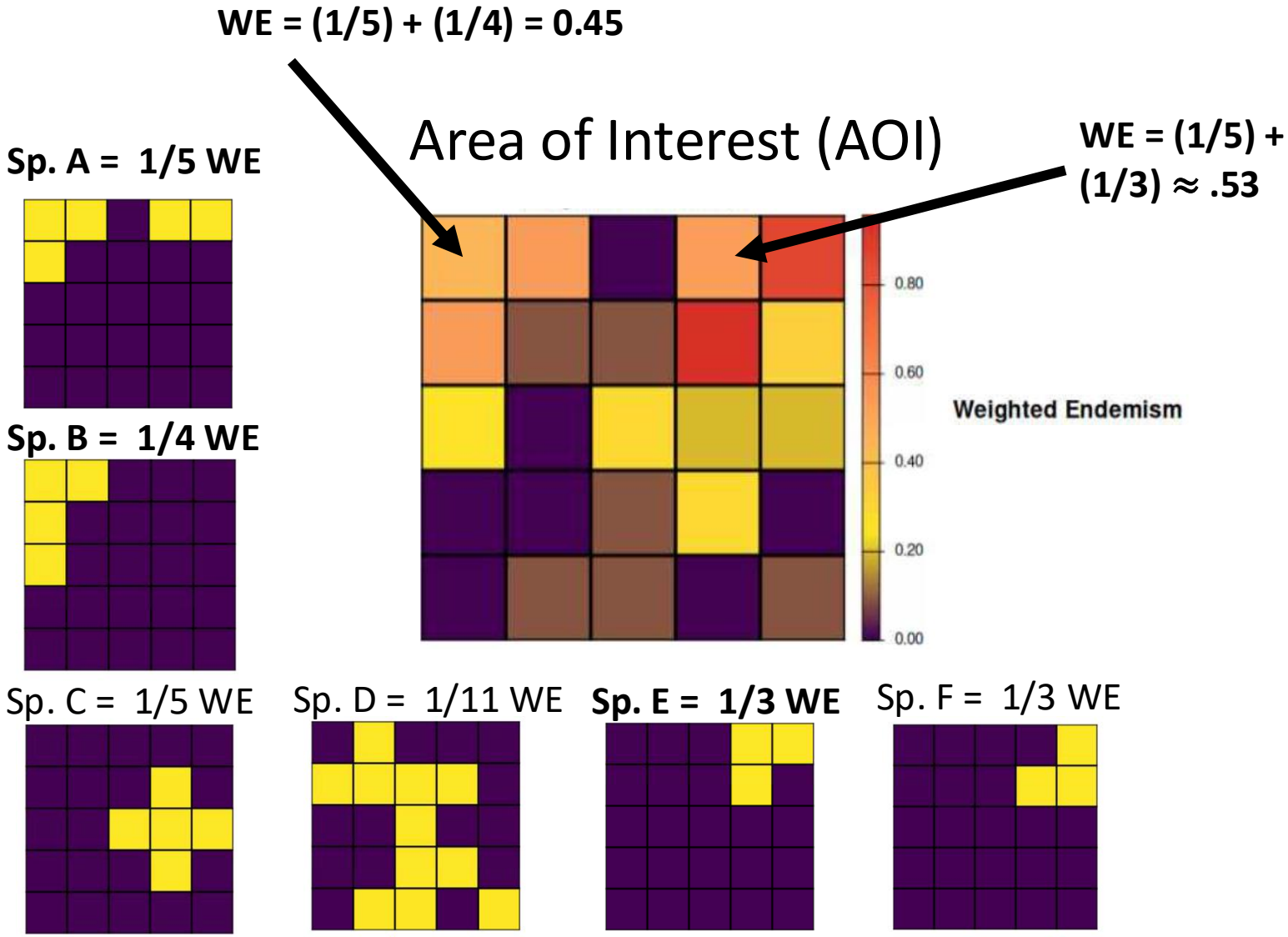
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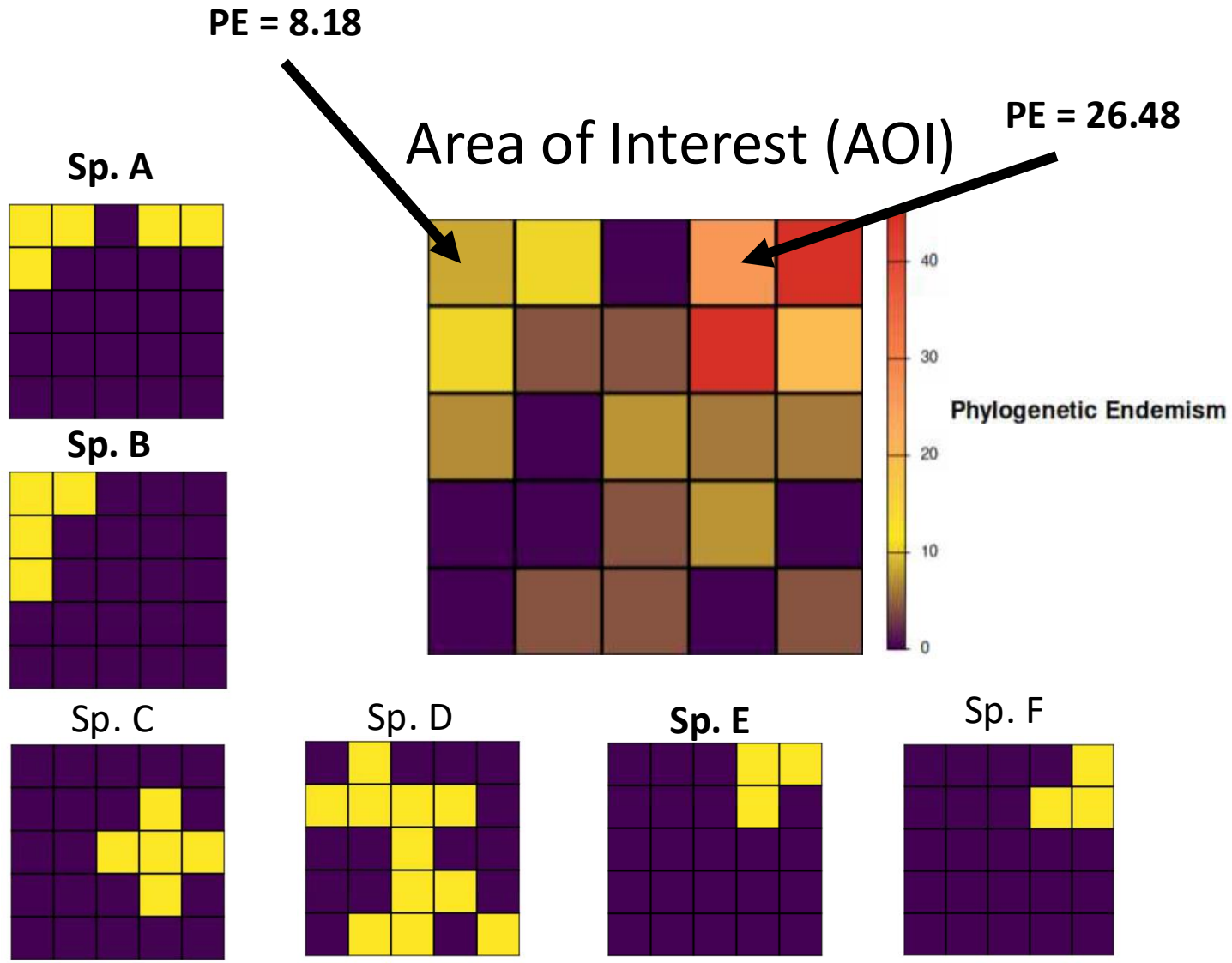
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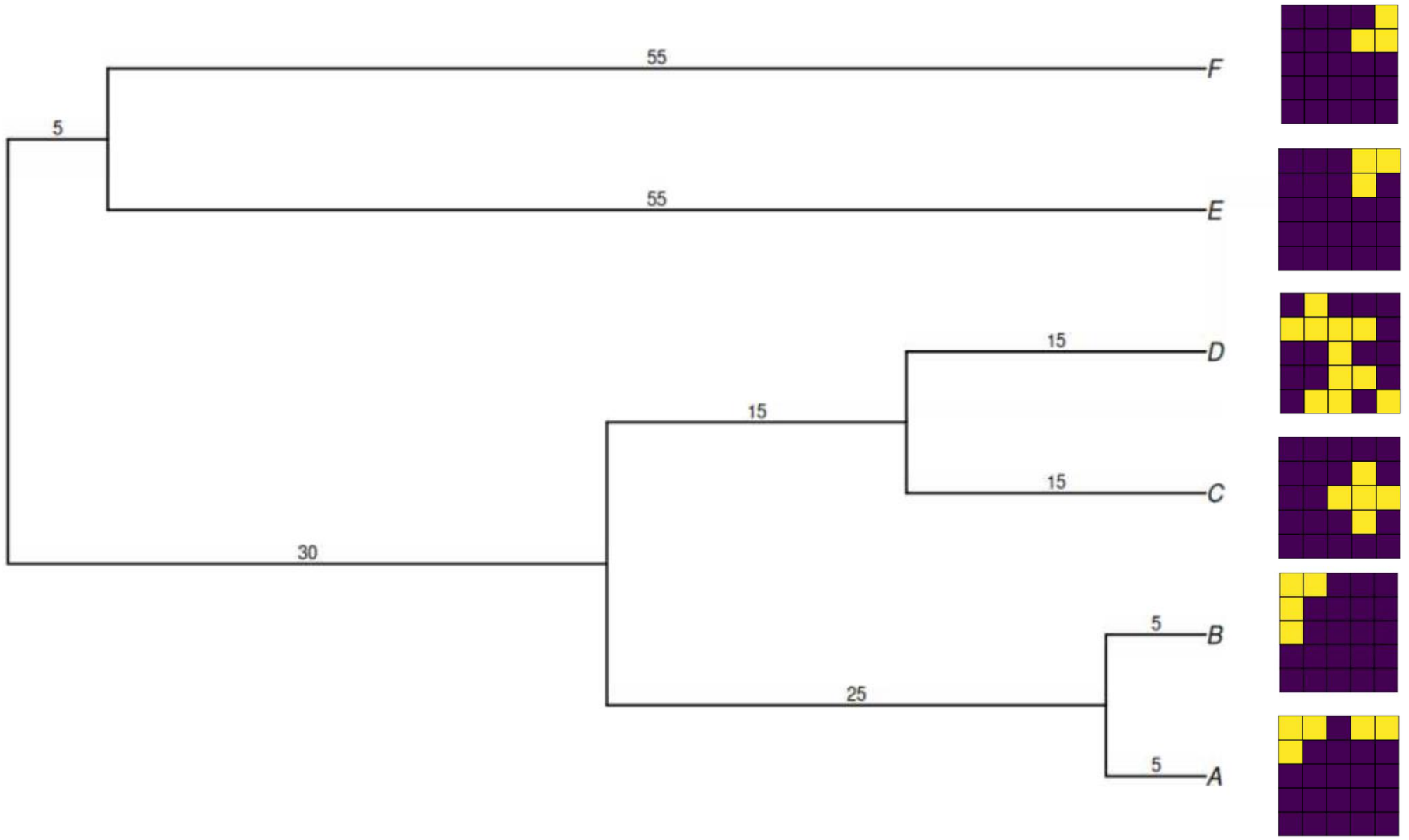
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(Weighted) Phylogenetic Endemism

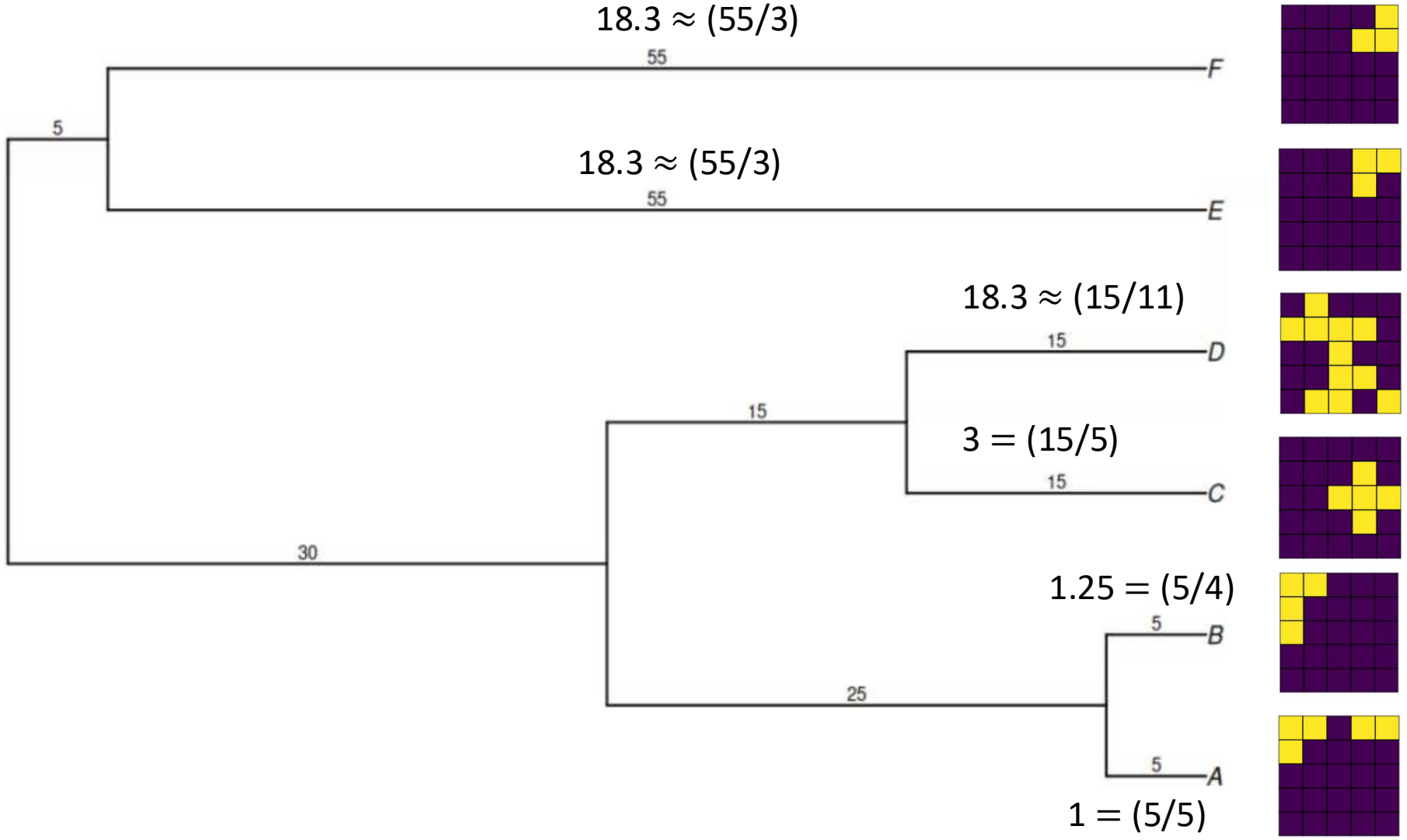
PE is PD on a range weighted tree



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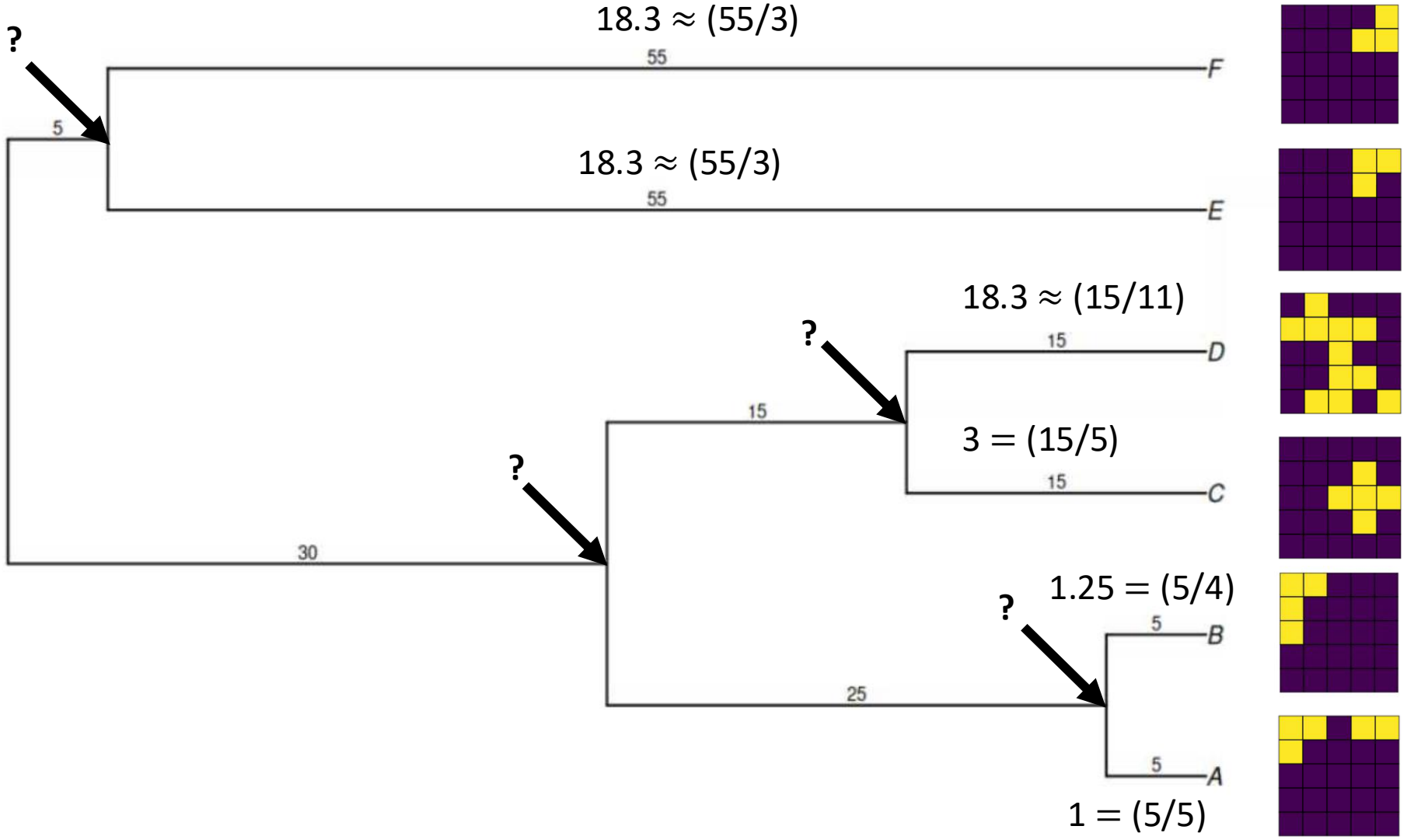
$$PE = \left(\frac{PD}{\#\ Presence\ Cells}\right)$$



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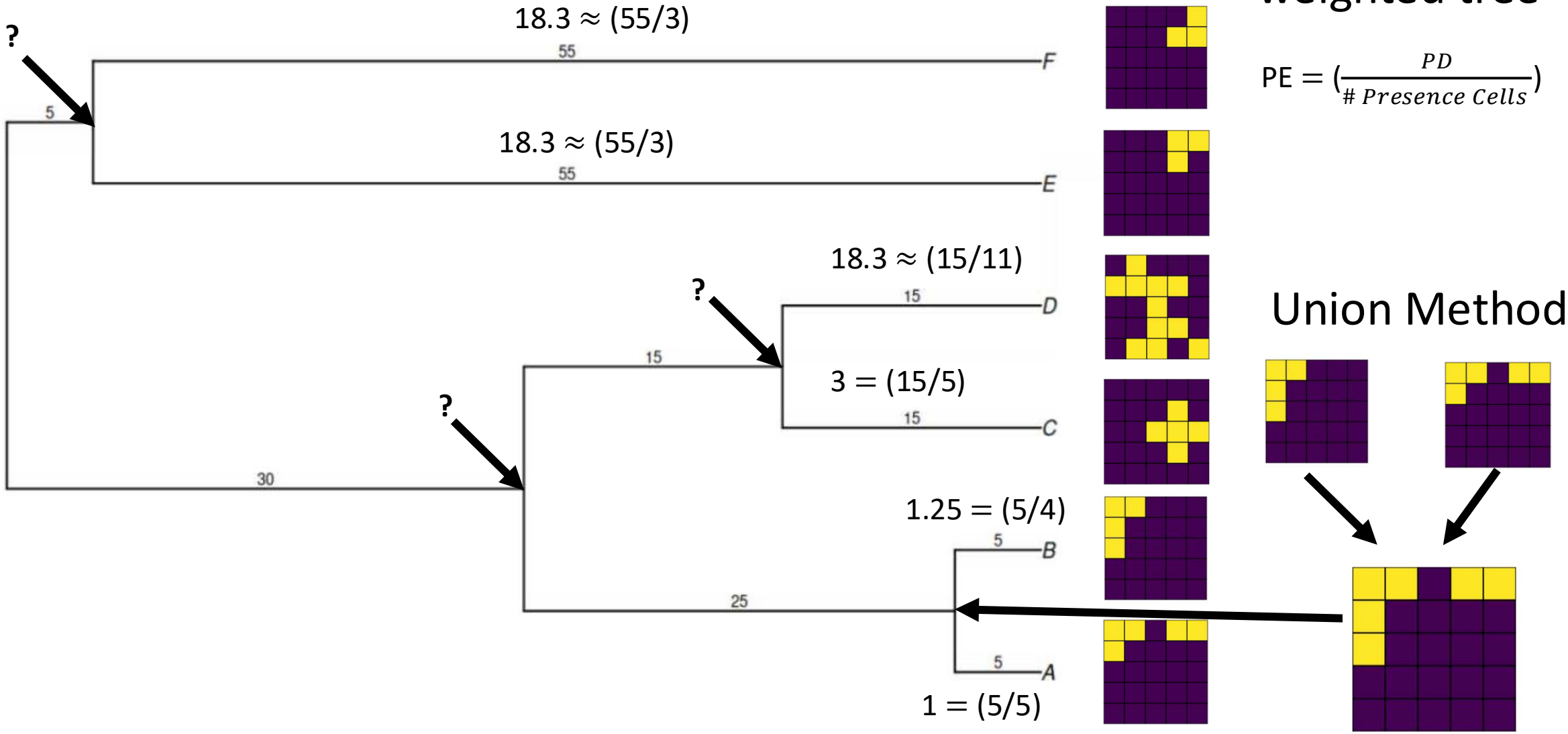
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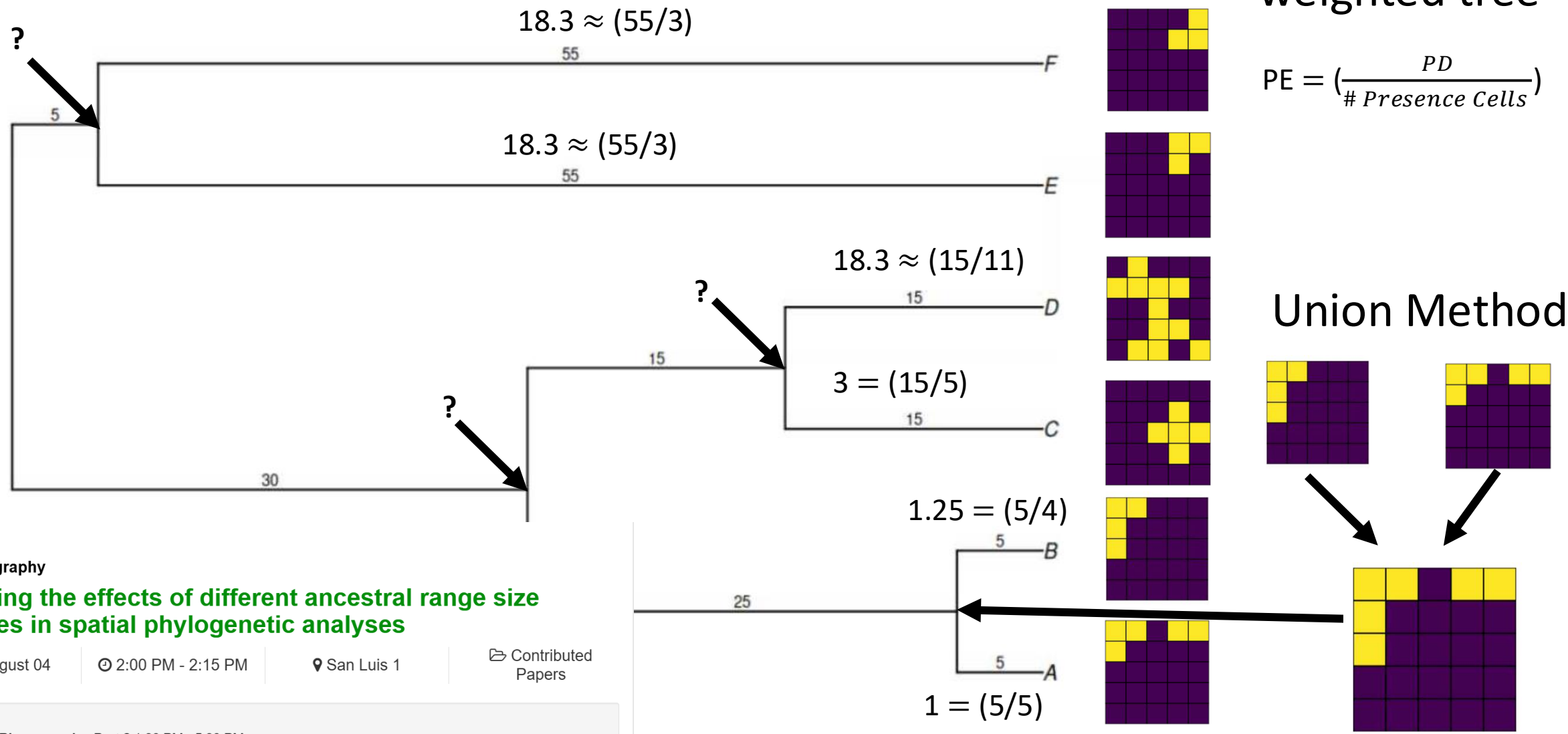
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(Weighted) Phylogenetic Endemism

PE is PD on a range weighted tree

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Union Method

Permalink

Biogeography

Assessing the effects of different ancestral range size estimates in spatial phylogenetic analyses

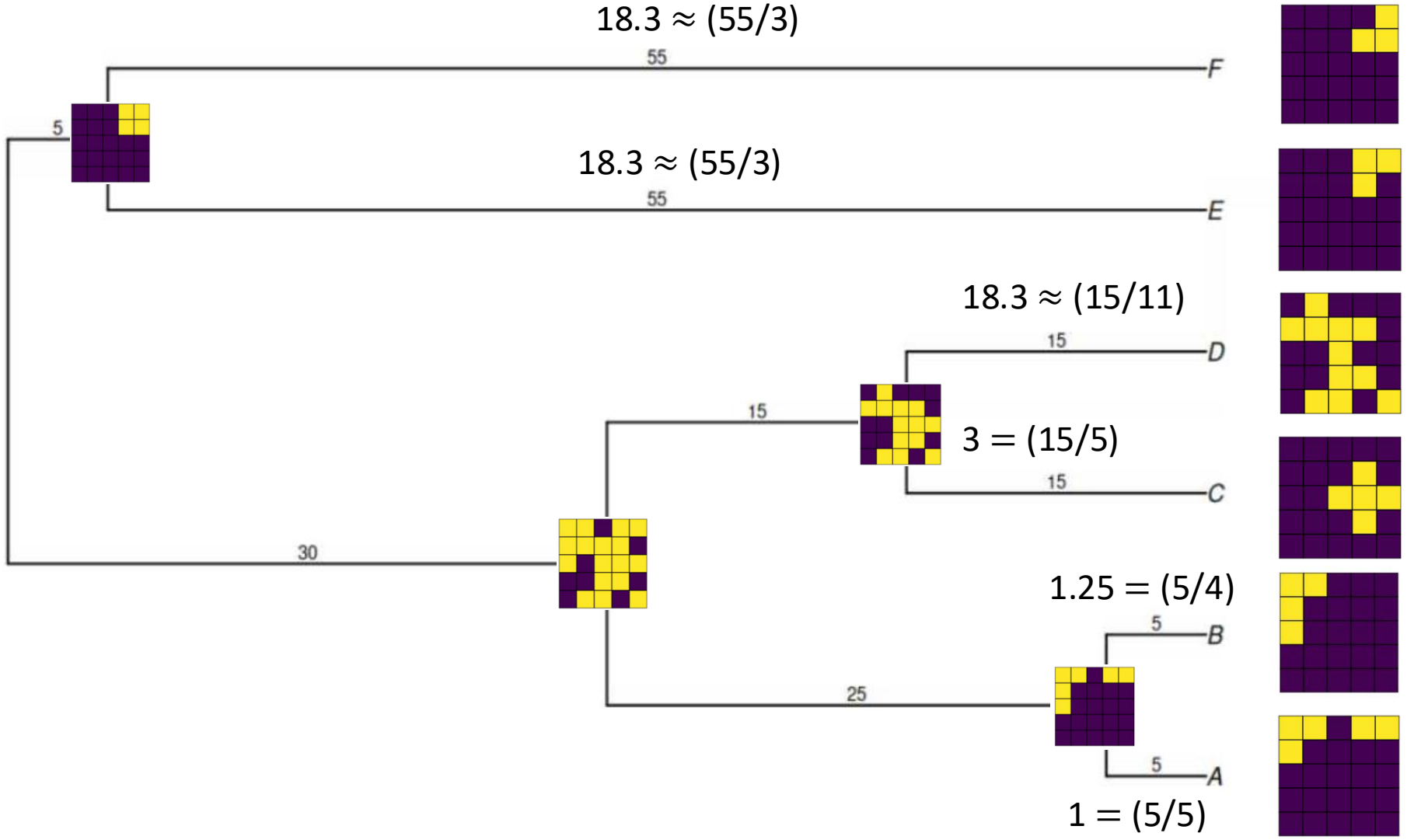
Tue, August 04 | 2:00 PM - 2:15 PM | San Luis 1 | Contributed Papers

Part of:
Session 30 - Biogeography, Part 2 1:30 PM - 5:30 PM

(Weighted) Phylogenetic Endemism

PE is PD on a range weighted tree

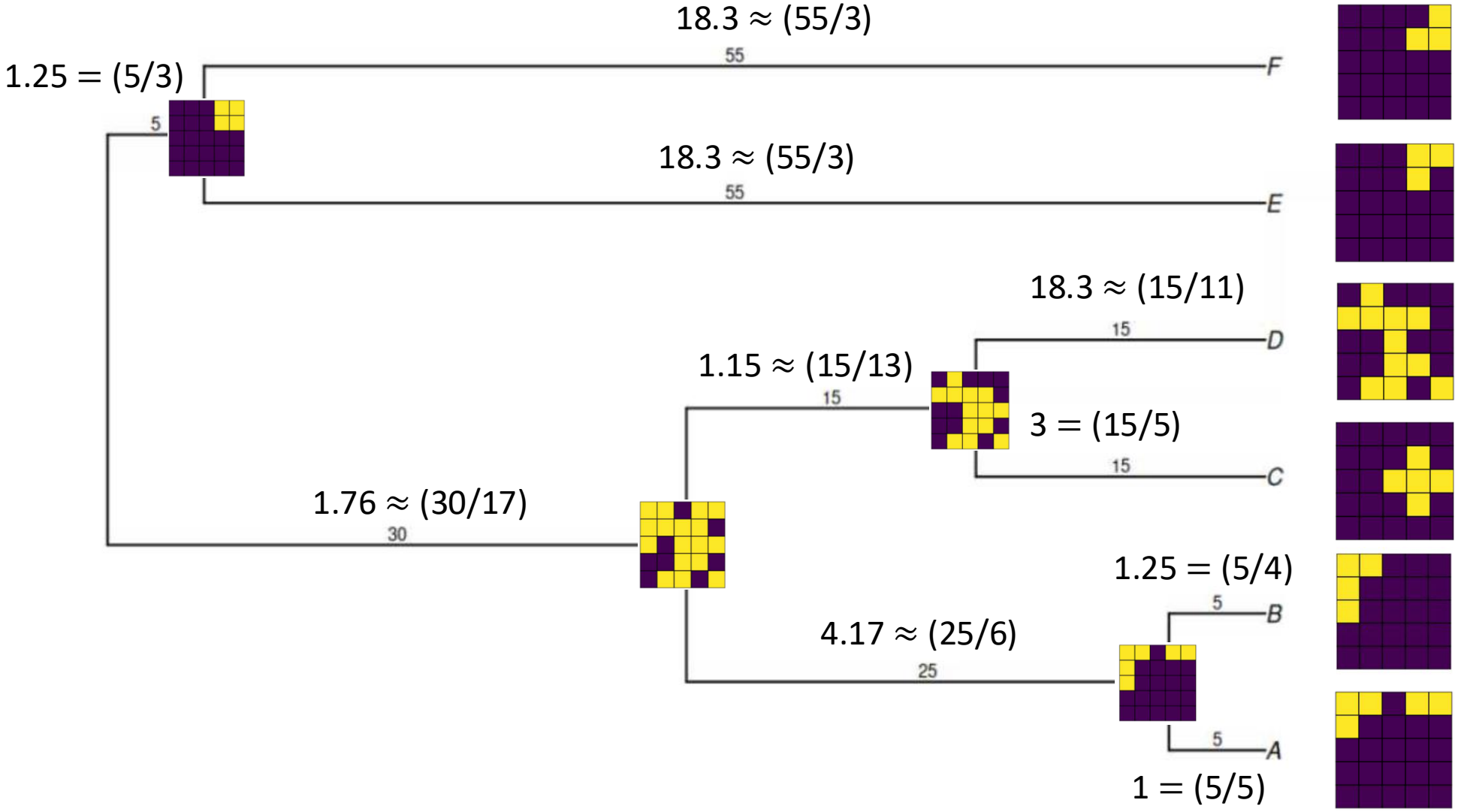
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(Weighted) Phylogenetic Endemism

PE is PD on a range weighted tree

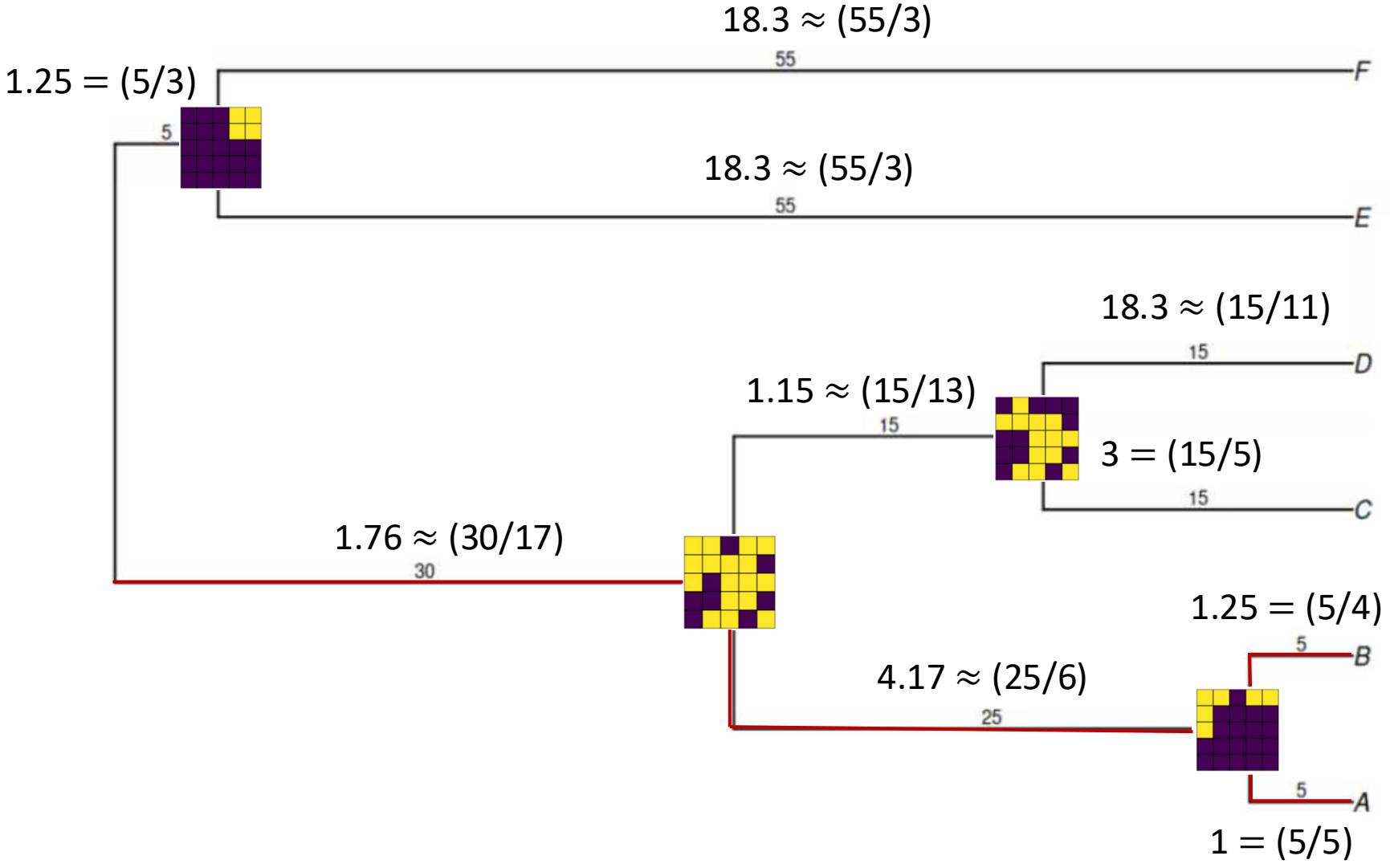
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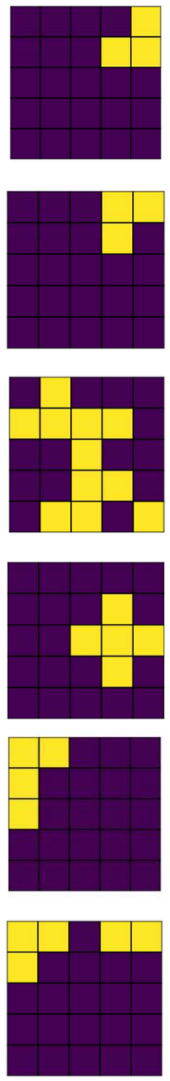


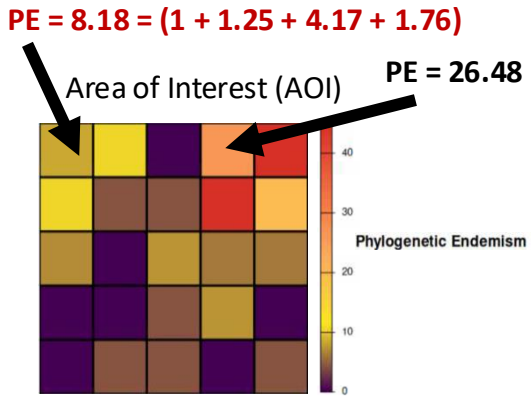
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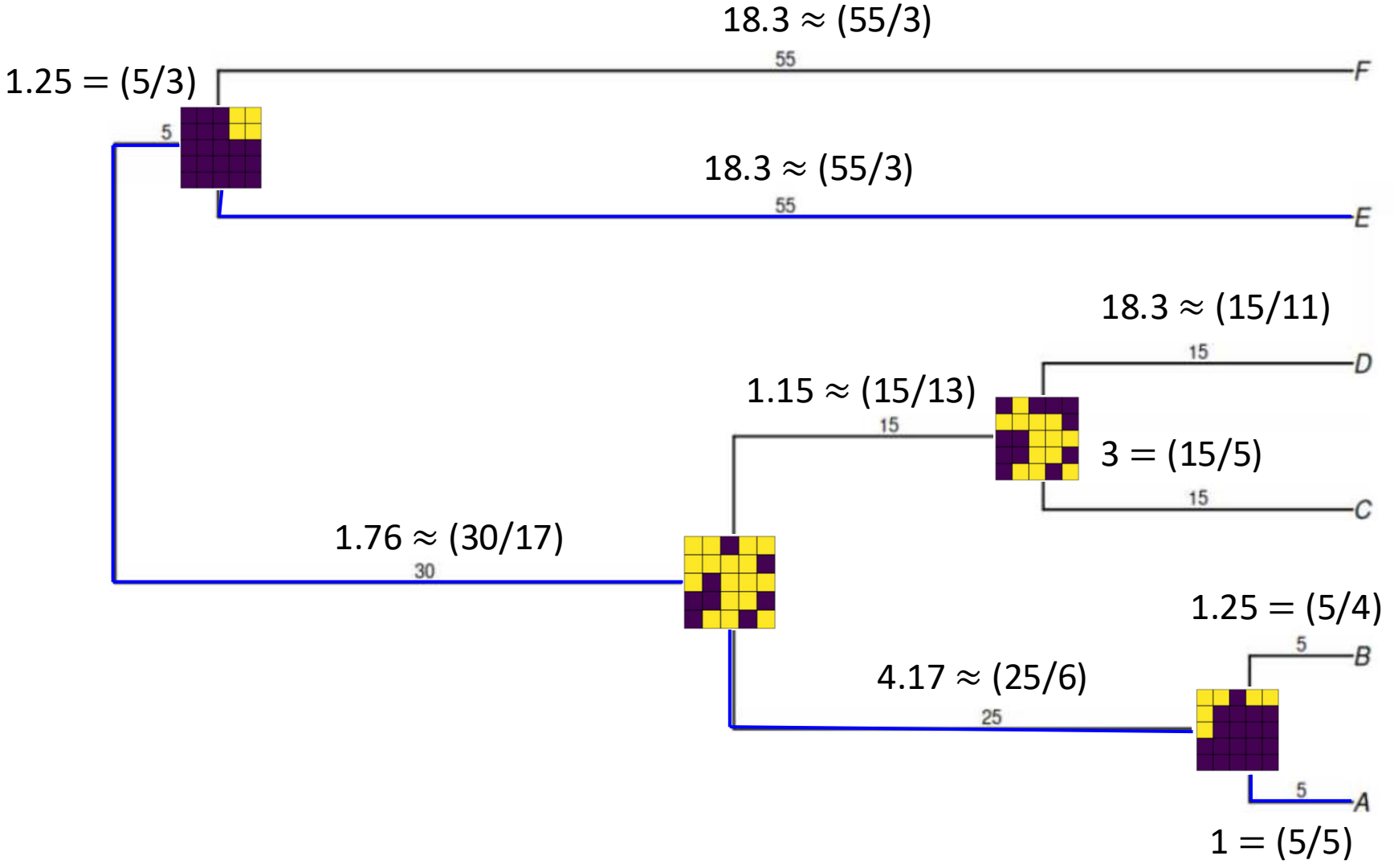


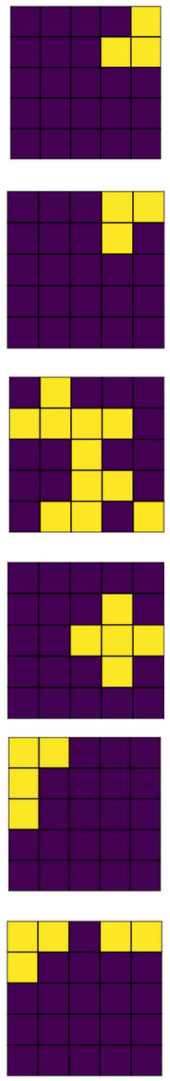


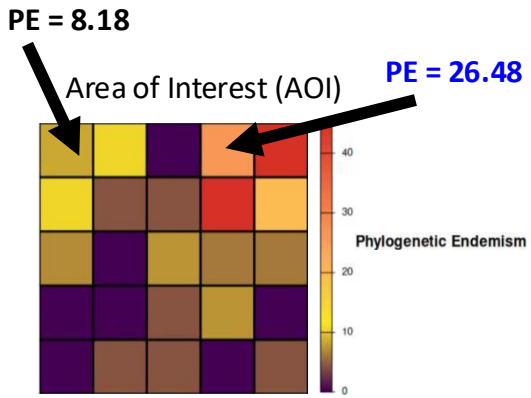
(Weighted) Phylogenetic Endemism

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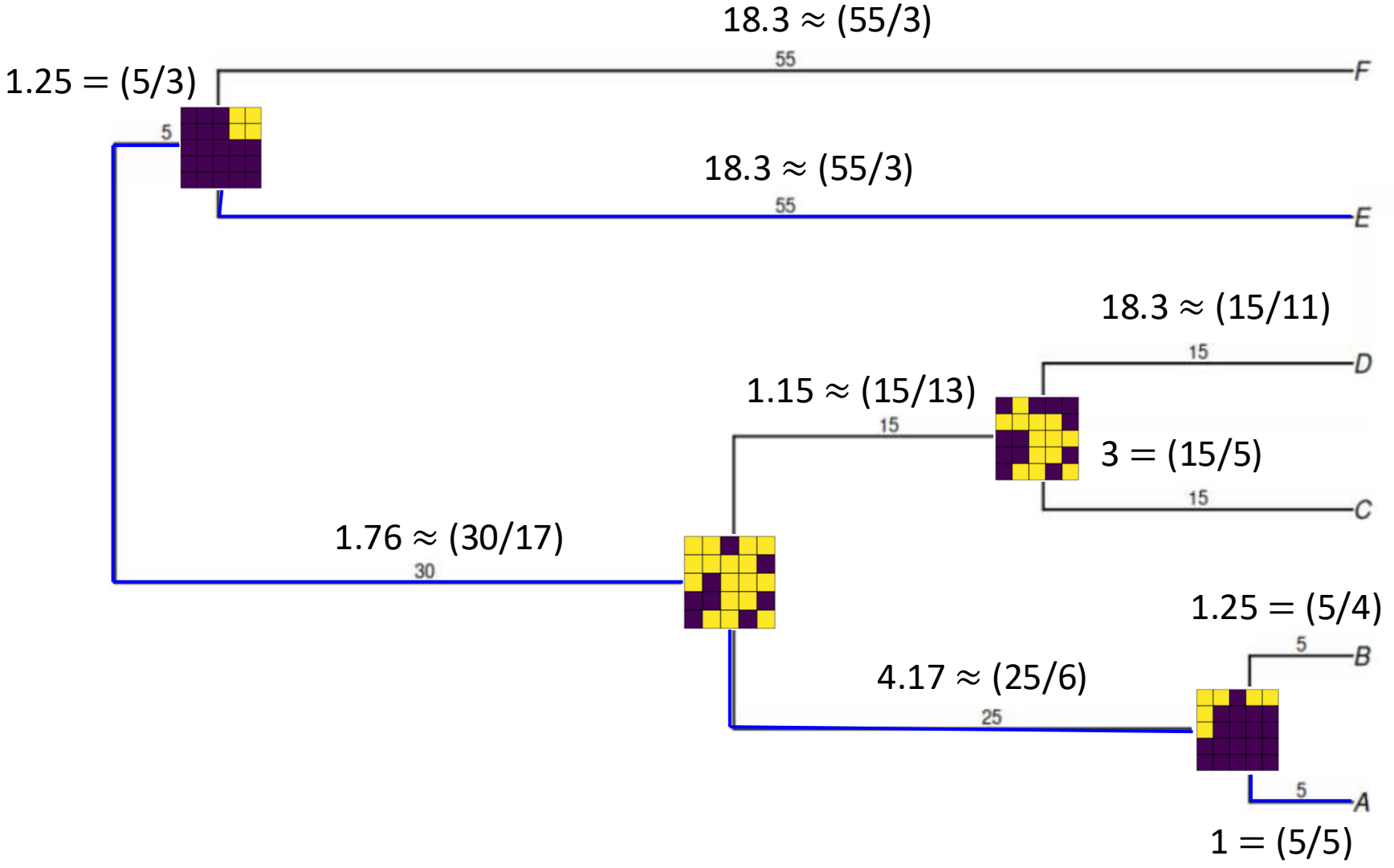


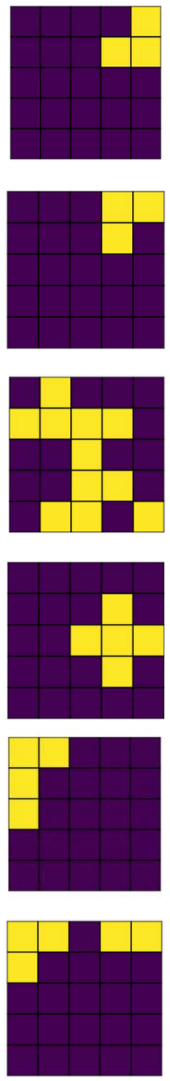


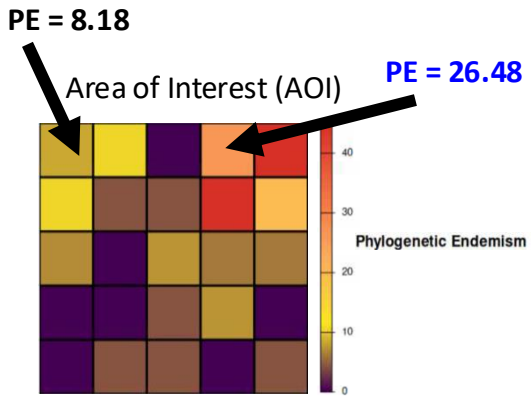
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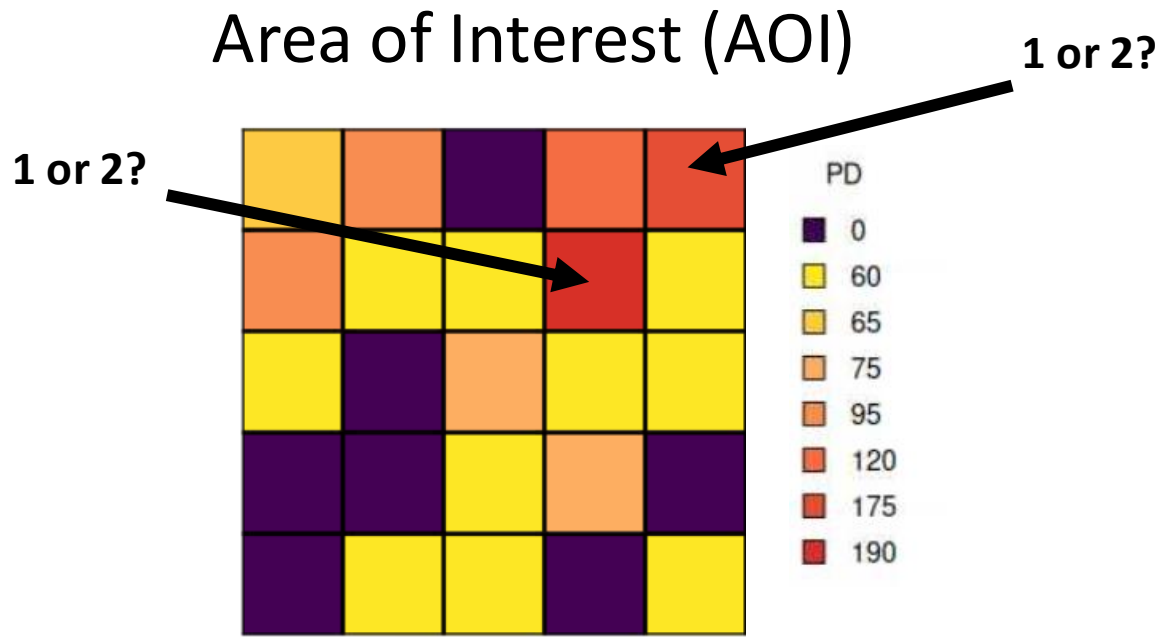


Intro to RPD and RPE

Currently an area with high PD/PE can be that way for two reasons

1) There is a lot of taxa (or narrow ranged taxa) in that cell
2) The cell has a disproportionately long or short branches

Relative **P**hylogenetic **D**iversity (RPD) and **R**elative **P**hylogenetic **E**ndemism (RPE) are designed to standardize PD and PE by comparing them to what would be expected if **all branches in the phylogeny were equal length** (aka a **comparison tree**)

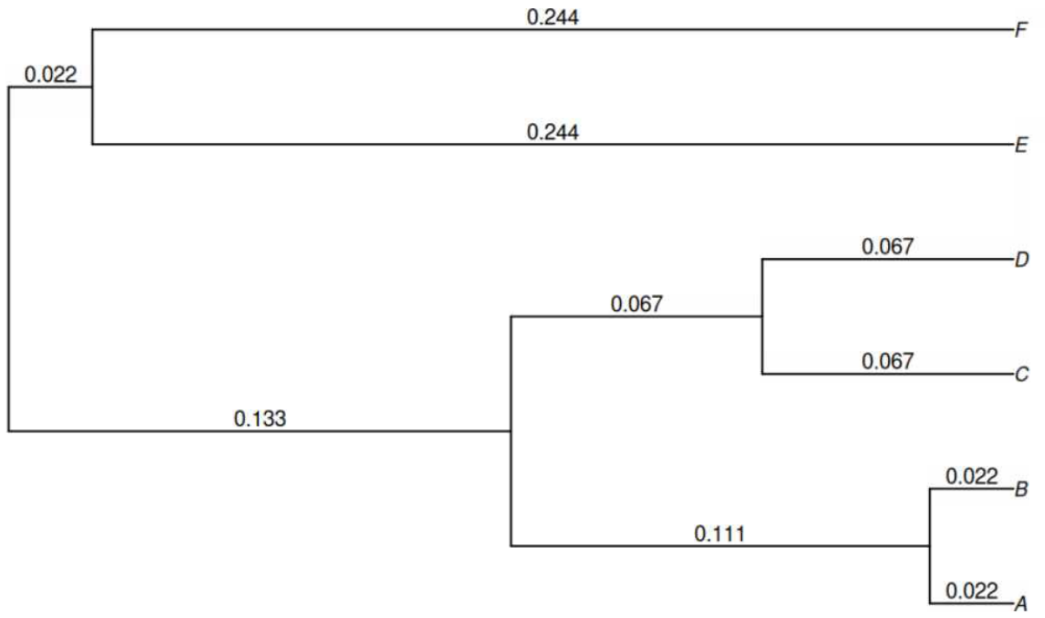


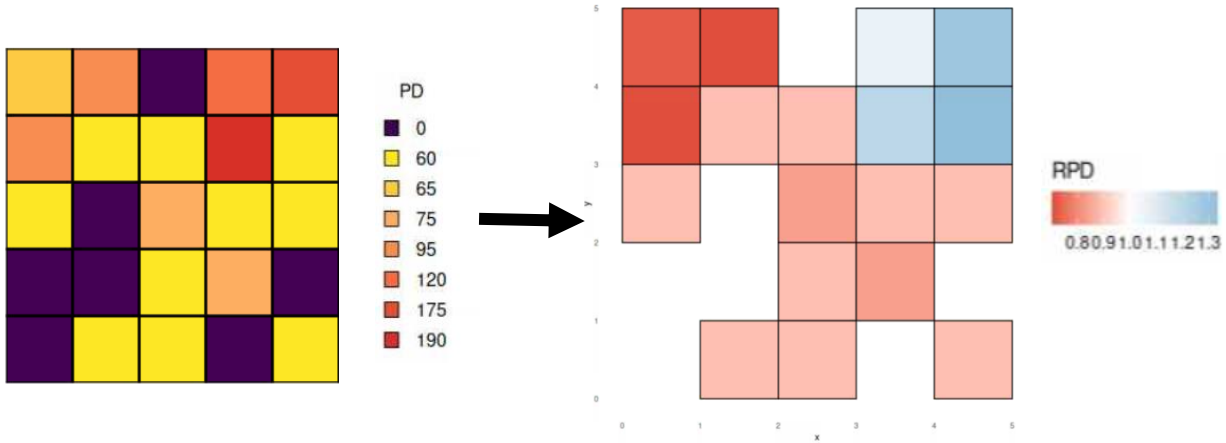
Intro to RPD and RPE

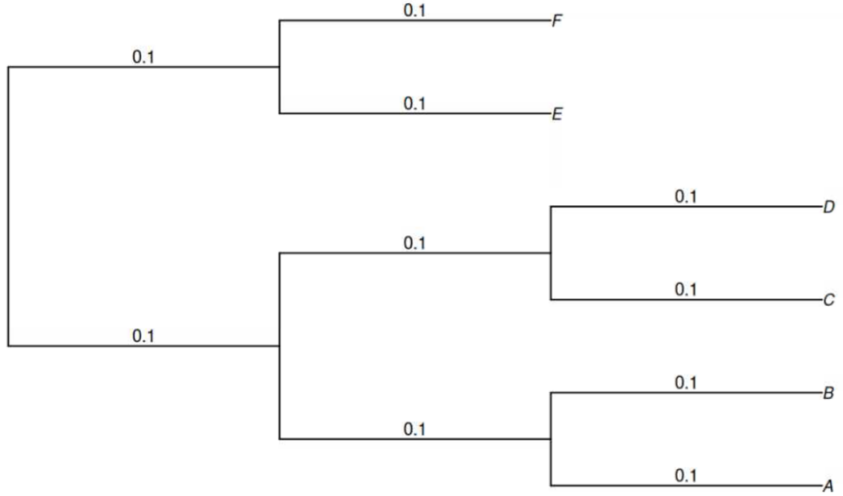
$$\text{RPD} = \left(\frac{PD\ observed\ tree}{PD\ comparison\ tree}\right)$$

RPD > 1 = Cell Contains **Longer** Branches than Expected

RPD < 1 = Cell Contains **Shorter** Branches than Expected





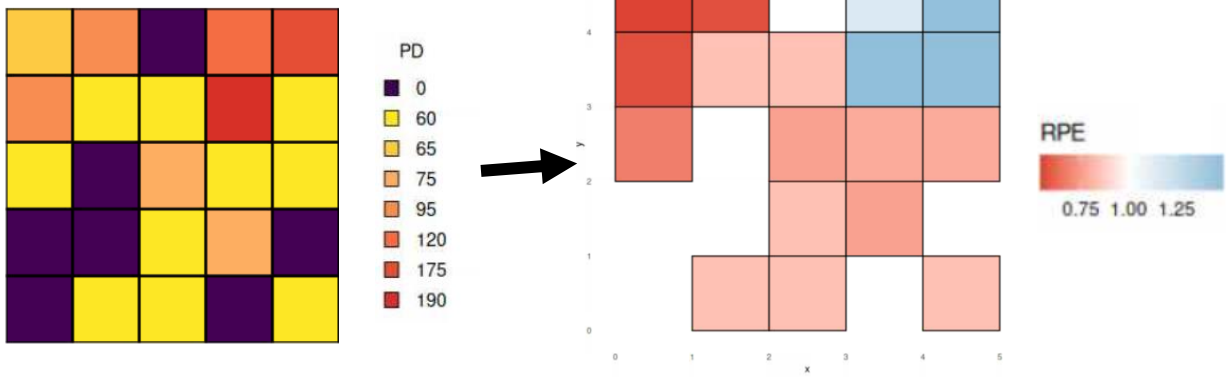


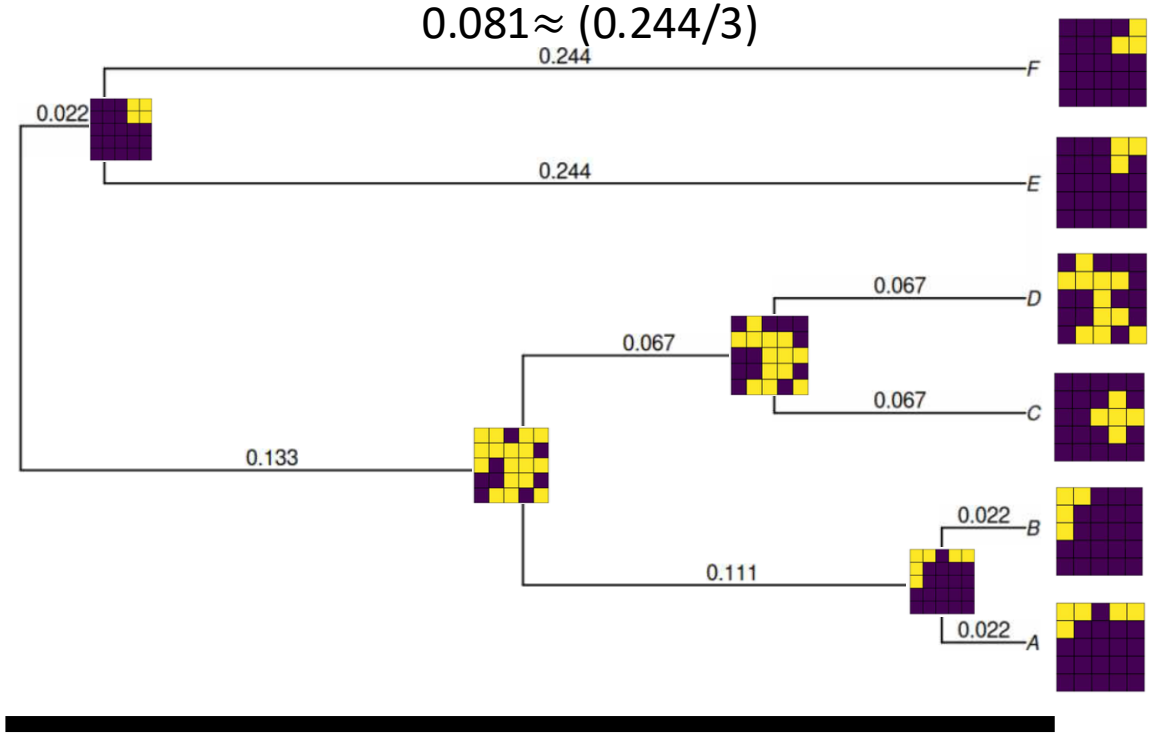
Intro to RPD and RPE

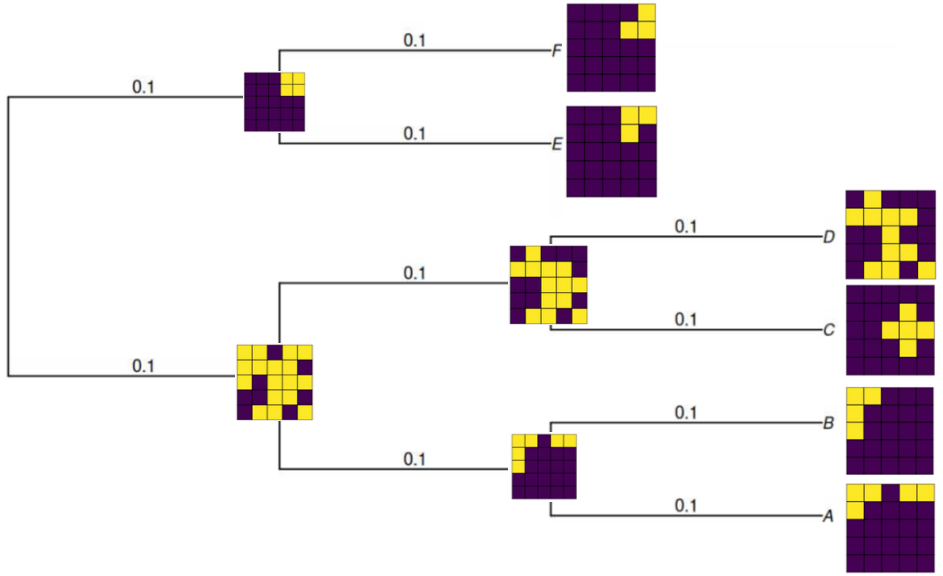
$$\text{RPE} = \left(\frac{PE\ observed\ tree}{PE\ comparison\ tree}\right)$$

RPE > 1 = Cell Contains **Range Restricted Longer** Branches than Expected

RPE < 1 = Cell Contains **Range Restricted Shorter** Branches than Expected

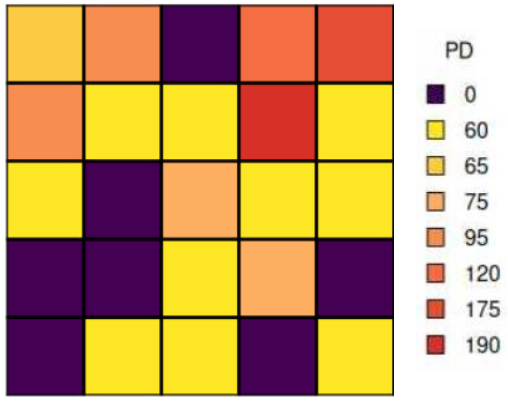




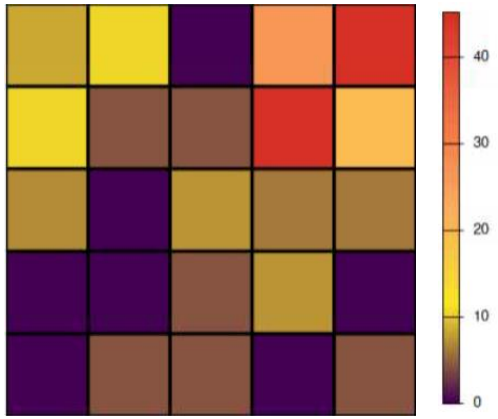


So Many Metrics…But what is random noise vs biology?

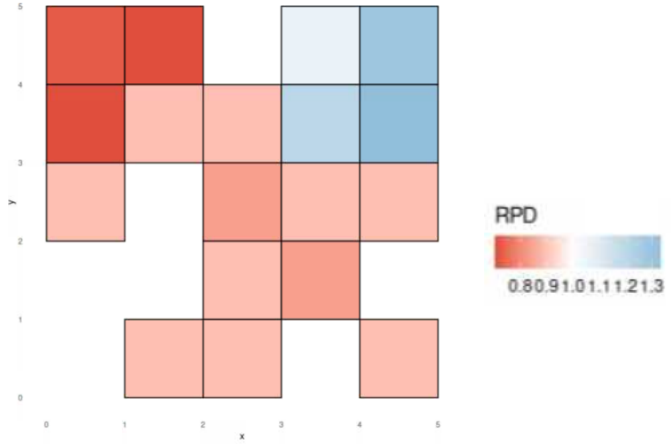
PD



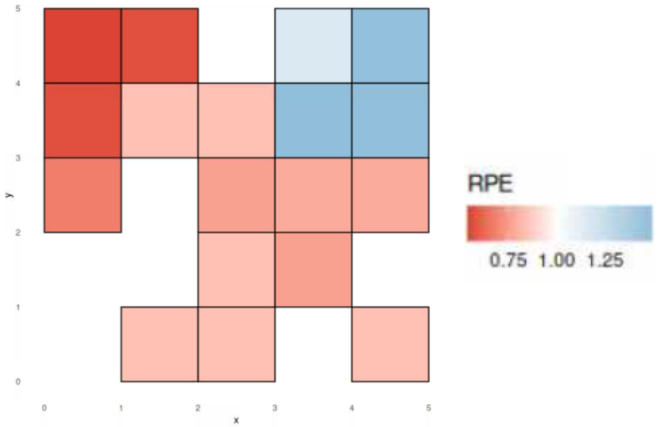
PE



RPD



RPE



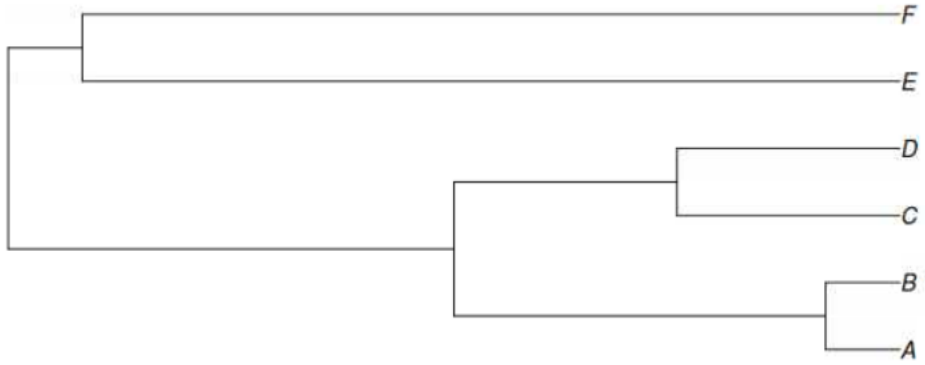
More Null Models!

Community Randomizations

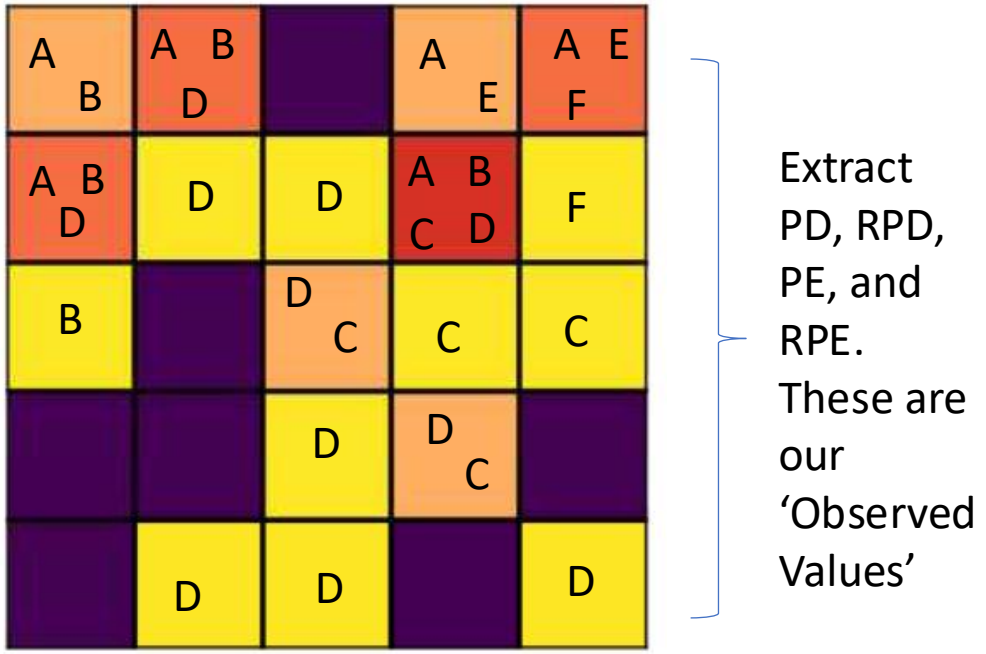
Community randomizations are a way to generate a null distribution (the expected values of community metrics under a specific null hypothesis)

"Curveball" algorithm

- What remains constant:
  - The Phylogenetic Tree Structure
  - Species Richness per Cell
  - Species Overall Prevalence (Total # Occurrences)
- What gets shuffled
  - The community matrix



Original Community Matrix



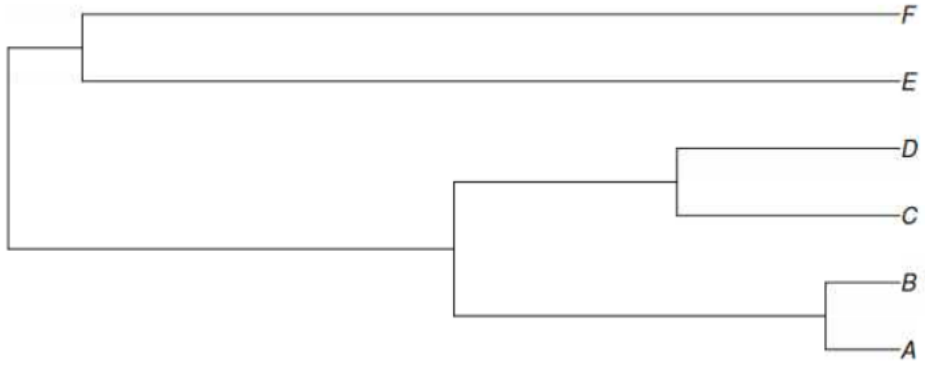
Extract PD, RPD, PE, and RPE. These are our 'Observed Values'

Community Randomizations

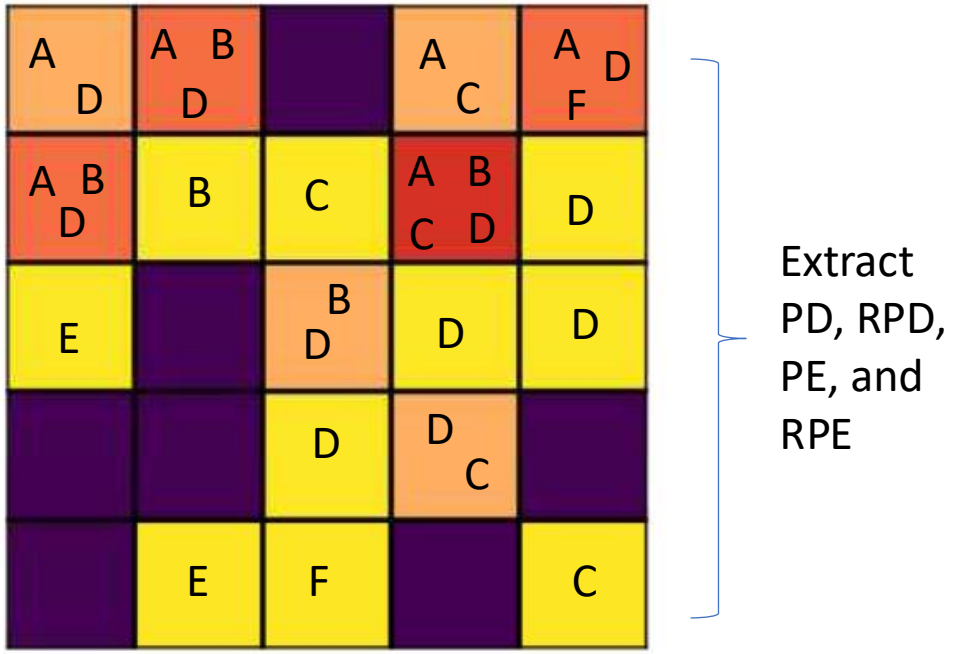
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  - The community matrix



Randomized Community Matrix x1

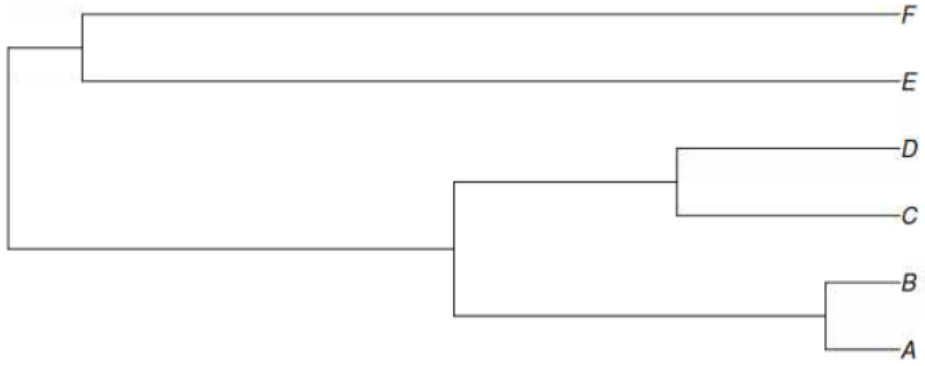


Community Randomizations

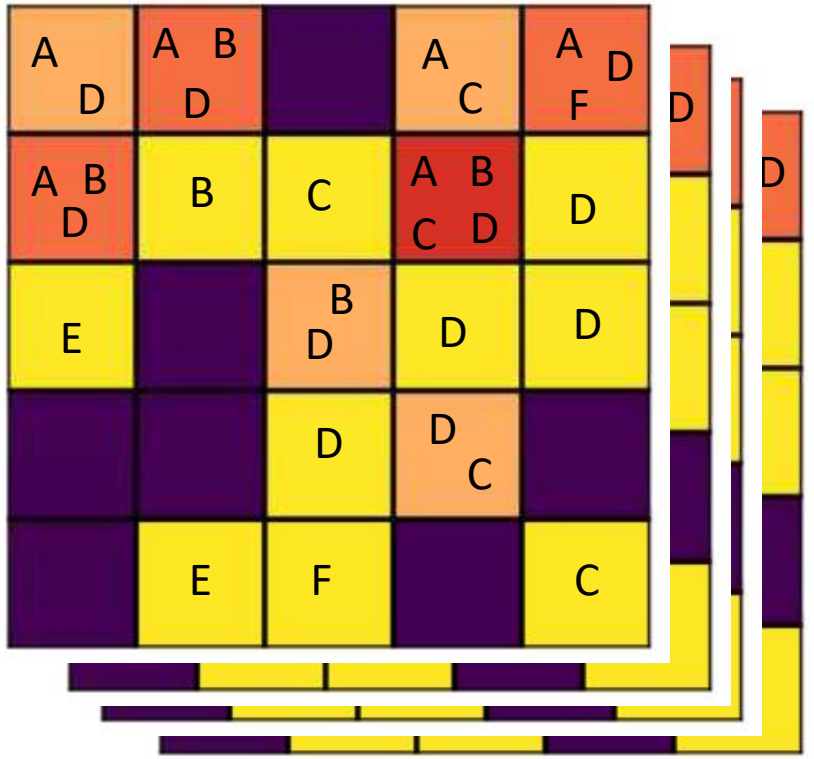
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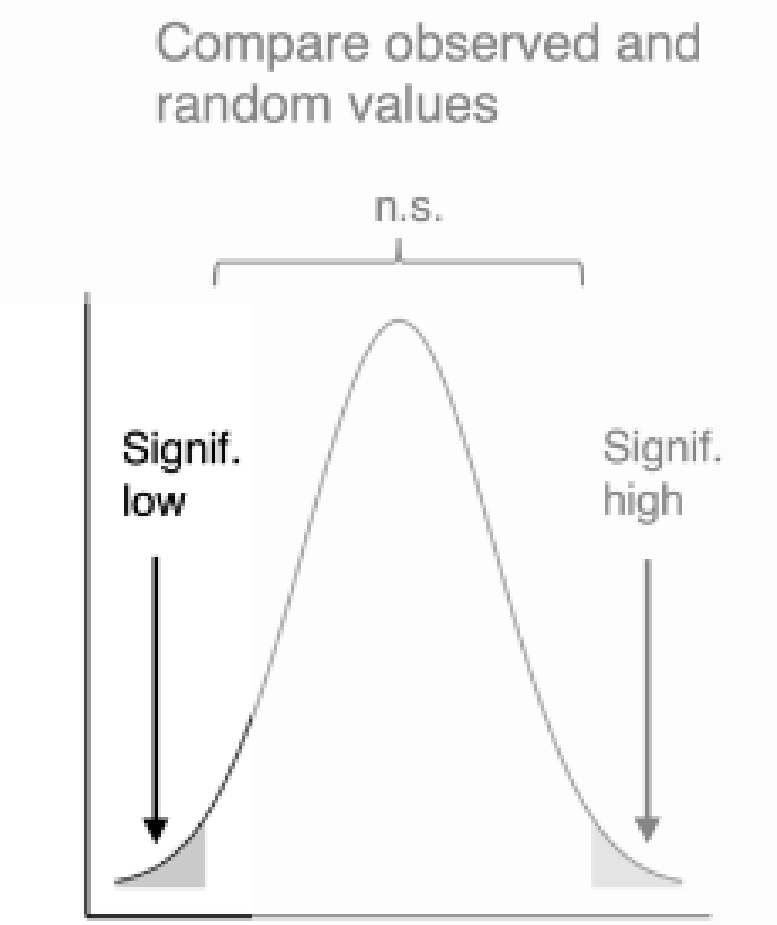
Randomized Community Matrix x999

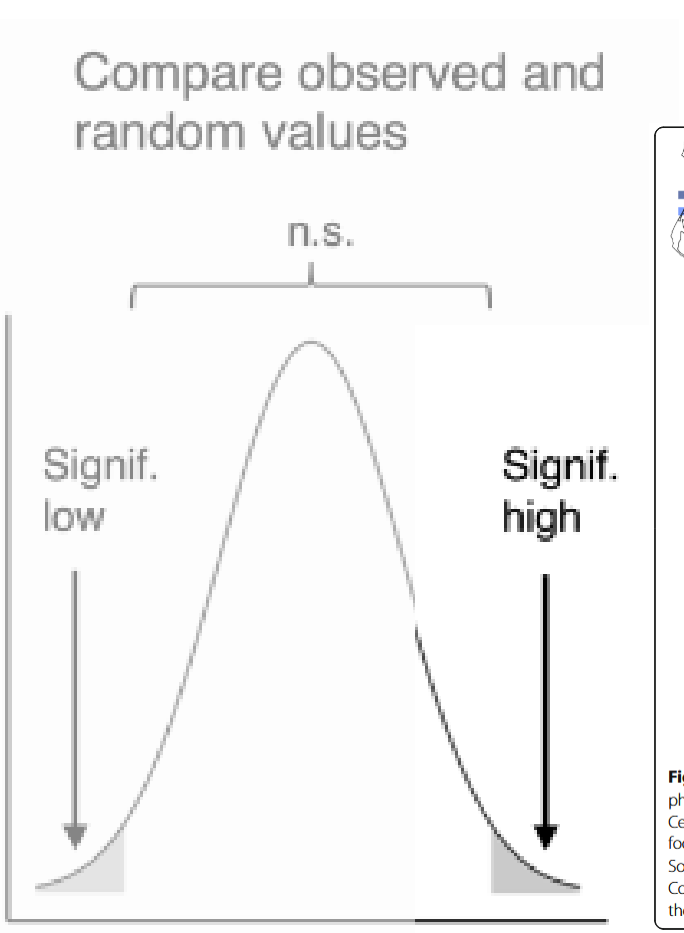


Repeat Extraction of PD, RPD, PE, and RPD to collect all the randomized null values per cell

Community Randomizations

Test whether your observed value is in either the lower or upper tail of the null distribution





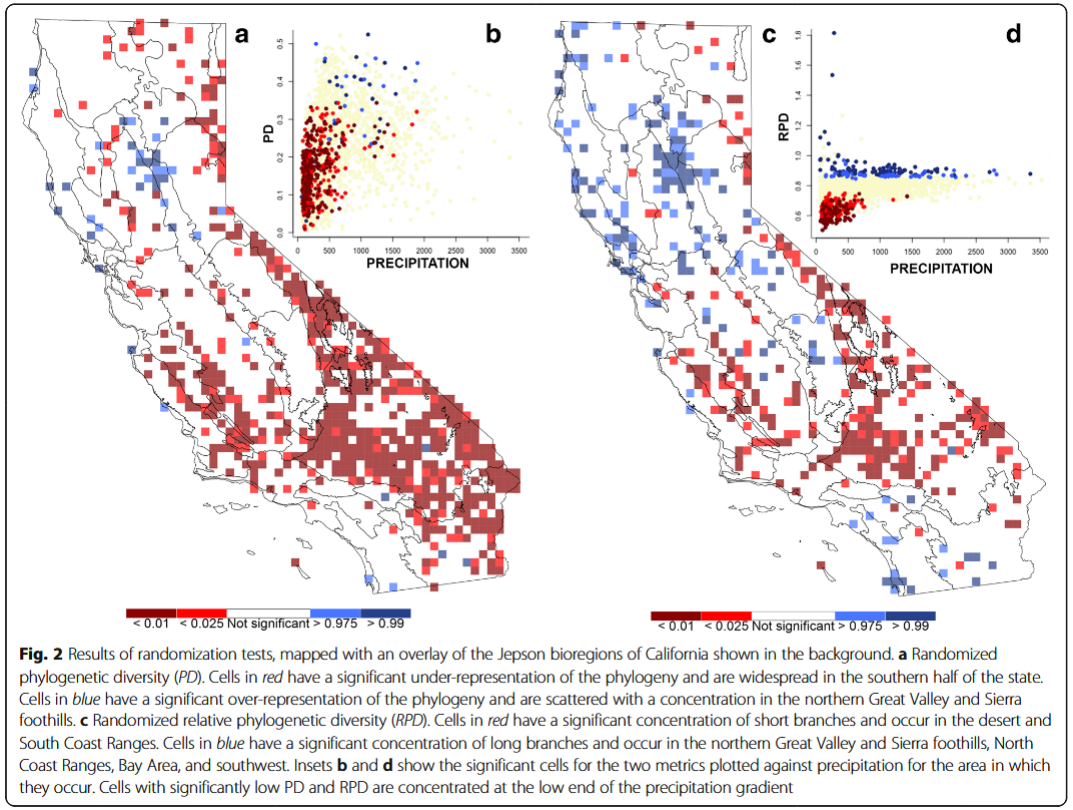


Fig. 2 Results of randomization tests, mapped with an overlay of the Jepson bioregions of California shown in the background. **a** Randomized phylogenetic diversity (*PD*). Cells in *red* have a significant under-representation of the phylogeny and are widespread in the southern half of the state. Cells in *blue* have a significant over-representation of the phylogeny and are scattered with a concentration in the northern Great Valley and Sierra foothills. **c** Randomized relative phylogenetic diversity (*RPD*). Cells in *red* have a significant concentration of short branches and occur in the desert and South Coast Ranges. Cells in *blue* have a significant concentration of long branches and occur in the northern Great Valley and Sierra foothills, North Coast Ranges, Bay Area, and southwest. Insets **b** and **d** show the significant cells for the two metrics plotted against precipitation for the area in which they occur. Cells with significantly low PD and RPD are concentrated at the low end of the precipitation gradient

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Categorization of Neo- and Paleo Endemism (CANAPE)

$$\text{Relative PE} = \frac{PE\ Observed\ Tree}{PE\ Comparison\ Tree}$$

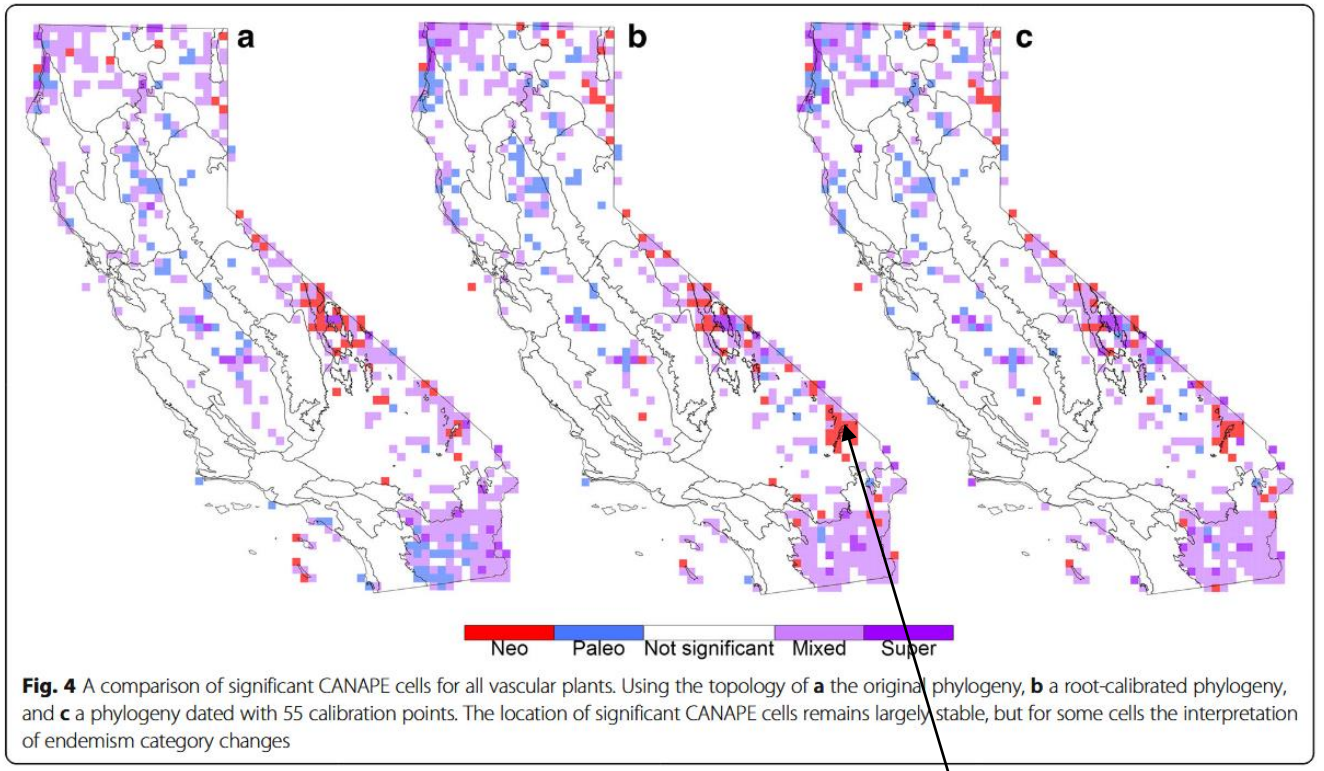
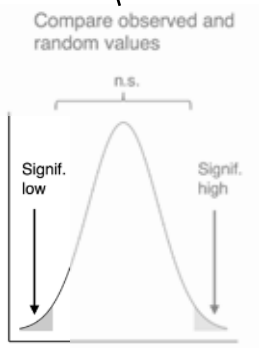


Fig. 4 A comparison of significant CANAPE cells for all vascular plants. Using the topology of **a** the original phylogeny, **b** a root-calibrated phylogeny, and **c** a phylogeny dated with 55 calibration points. The location of significant CANAPE cells remains largely stable, but for some cells the interpretation of endemism category changes

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CANAPE (Mishler et al. 2014)

1. Is a cell significant for RPE? → 2. Two-tailed Test on RPE ratio



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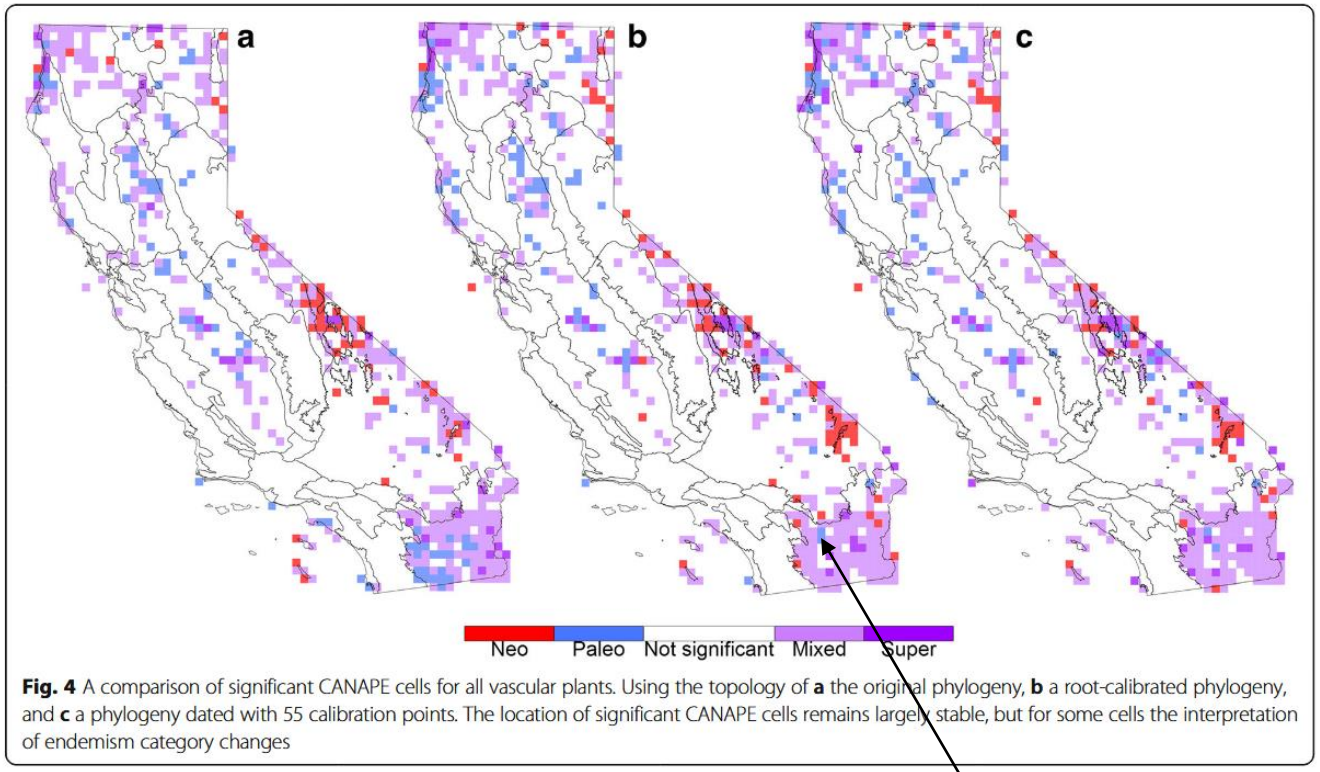
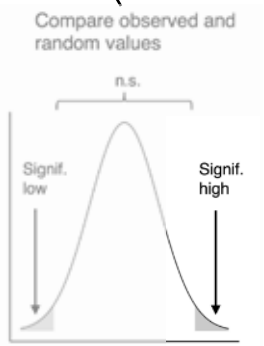


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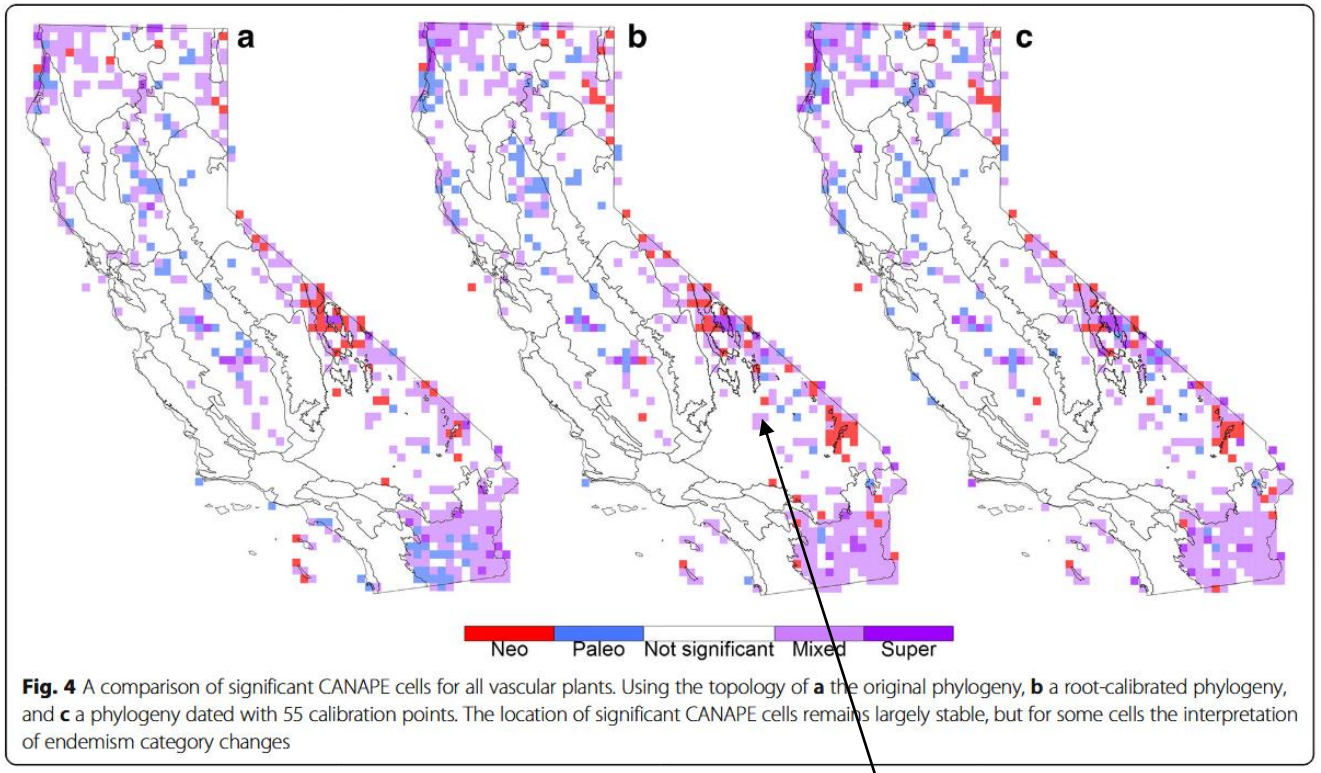
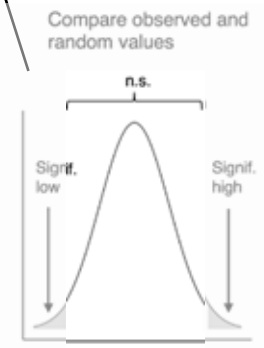


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Thornhill et al. 2017

CANAPE (Mishler et al. 2014)

1. Is a cell significant for RPE? → 2. Two-tailed Test on RPE ratio



Time for a well-deserved Coffee Break. Then we'll look at applying these methods to the Sonoran Basin and Range's Cacti





Western Pricklypear (*Opuntia orbiculate)*